

模拟电子技术基础

Analog Device

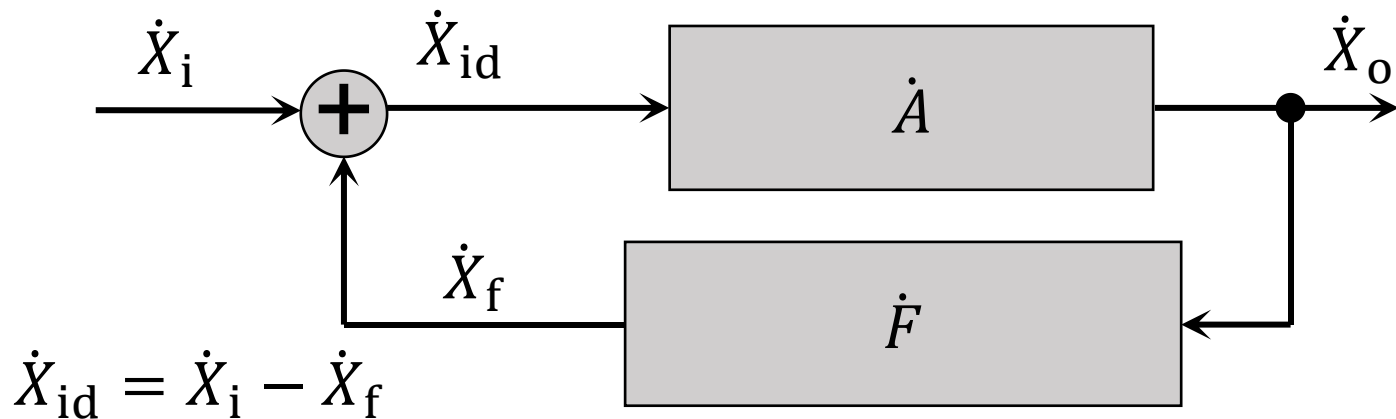
6 Feedback amplifier circuit

反馈放大电路

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6.3 The effect of negative feedback on amplifier circuit



Open-loop gain 开环增益 $\dot{A} = \frac{\dot{X}_o}{\dot{X}_{id}}$

Close-loop gain 闭环增益
Gain with feedback 反馈增益 $\dot{A}_f = \frac{\dot{X}_o}{\dot{X}_i}$

Feedback factor 反馈系数 $\dot{F} = \frac{\dot{X}_f}{\dot{X}_o}$

Loop gain 环路增益 $\dot{A}\dot{F} = \frac{\dot{X}_f}{\dot{X}_{id}}$

Close-loop gain 闭环增益 $\dot{A}_f = \frac{\dot{X}_o}{\dot{X}_i}$

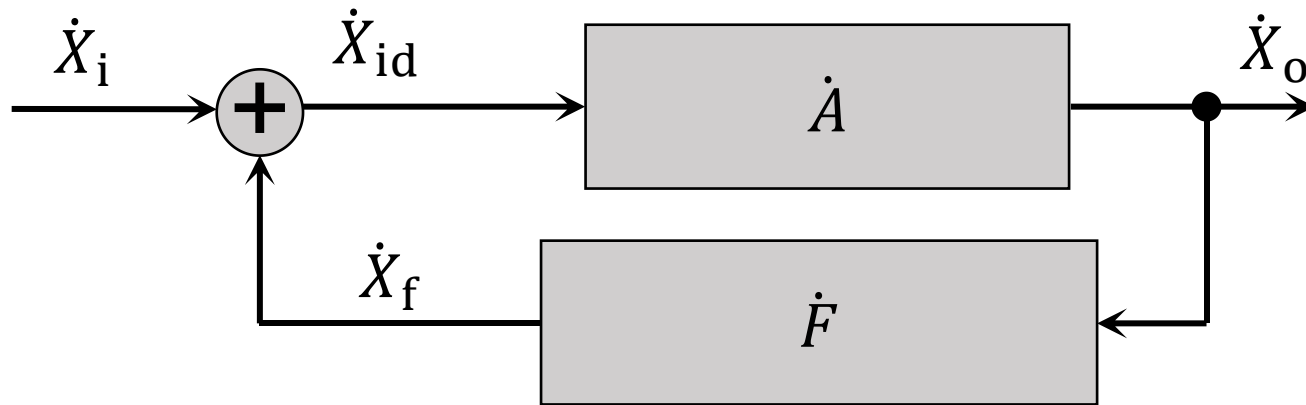
$$= \frac{\dot{X}_o}{\dot{X}_{id} + \dot{X}_f} = \frac{\dot{A}\dot{X}_{id}}{\dot{X}_{id} + \dot{A}\dot{F}\dot{X}_{id}}$$

$$= \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$

In midband, $\dot{A}$, $\dot{F}$, and $\dot{A}_f$ are real number, hence

$$A_f = \frac{A}{1 + AF}$$

$1 + \dot{A}\dot{F}$ is called the **amount of feedback** 反馈深度



No feedback circuit:

$$\dot{A}_f = \dot{A}$$

With feedback circuit:

$$\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$

This is because the net input signal for amplification:

$$\dot{X}_{id} = \frac{\dot{X}_i}{1 + \dot{A}\dot{F}}$$

Close-loop gain 闭环增益

$$\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$

(1) When $|1 + \dot{A}\dot{F}| = 1$, $|\dot{A}_f| = |\dot{A}|$: no feedback

(2) When $|1 + \dot{A}\dot{F}| > 1$, $|\dot{A}_f| < |\dot{A}|$: negative feedback

When $|\dot{A}\dot{F}| \gg 1$,

$$\dot{A}_f \approx \frac{1}{\dot{F}}$$

Strong negative feedback 深度负反馈

Strong negative feedback 深度负反馈

When $|\dot{A}\dot{F}| \gg 1$,

$$\dot{A}_f \approx \frac{1}{\dot{F}}$$

- The close-loop gain only depends on the feedback network.
- Since feedback network is usually made of resistors and capacitors, the amplifier circuit is very stable.

Close-loop gain 闭环增益

$$\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$

(3) When $|1 + \dot{A}\dot{F}| < 1$, $|\dot{A}_f| > |\dot{A}|$: positive feedback

When $|1 + \dot{A}\dot{F}| = 0$:

There is output signal when no input signal.

This is called **self-excited oscillation** 自激振荡

Q1: There is a feedback amplifier circuit with open-loop gain of $A=300$ and feedback factor of $F=0.01$, ask:

(1) The close-loop gain A_f

$$\begin{aligned} A_f &= \frac{A}{1 + AF} \\ &= \frac{300}{1 + 300 \times 0.01} \\ &= 75 \end{aligned}$$

Q1: There is a feedback amplifier circuit with open-loop gain of $A=300$ and feedback factor of $F=0.01$, ask:

(2) When A increase 20%, A_f increase ?%

$$A' = (1 + 20\%) \times 300 = 360$$

$$A_f' = \frac{A'}{1 + A'F}$$

$$= \frac{360}{1 + 360 \times 0.01} = 78.26$$

$$\frac{\Delta A_f'}{A_f} = \frac{A_f' - A_f}{A_f'} = \frac{78.26 - 75}{75} = 4.3\%$$

Bandwidth of negative feedback circuit: high- f region

No negative feedback: $\dot{A}_h = \dot{A}_m \frac{1}{1 + j f / f_H}$

With negative feedback:

$$\begin{aligned}\dot{A}_{hf} &= \frac{\dot{A}_h}{1 + \dot{A}_h \dot{F}} = \frac{\dot{A}_m / (1 + j \frac{f}{f_H})}{1 + \dot{A}_m \dot{F} / (1 + j \frac{f}{f_H})} \\ &= \frac{\dot{A}_m / (1 + \dot{A}_m \dot{F}_h)}{1 + j \frac{f}{(1 + \dot{A}_m \dot{F}_h) f_H}} = \frac{\dot{A}_{mf}}{1 + j \frac{f}{f_{Hf}}}\end{aligned}$$

$$\dot{A}_{mf} = \frac{\dot{A}_m}{(1 + \dot{A}_m \dot{F}_h)}$$

$$f_{Hf} = (1 + \dot{A}_m \dot{F}_h) f_H$$

Bandwidth of negative feedback circuit: low- f region

No negative feedback: $\dot{A}_l = \dot{A}_m \frac{j f / f_L}{1 + j f / f_L}$

With negative feedback: $\dot{A}_{lf} = \dot{A}_{mf} \frac{j \frac{f}{f_{Lf}}}{1 + j \frac{f}{f_{Lf}}}$

$$\dot{A}_{mf} = \frac{\dot{A}_m}{(1 + \dot{A}_m \dot{F}_h)}$$

$$f_{Lf} = \frac{f_L}{(1 + \dot{A}_m \dot{F}_h)}$$

Bandwidth of negative feedback circuit

$$f_{\text{bwf}} = f_{\text{Hf}} - f_{\text{Lf}} \approx f_{\text{Hf}} = (1 + \dot{A}_m \dot{F}_h) f_{\text{H}}$$

Gain in midband

$$\dot{A}_{\text{mf}} = \frac{\dot{A}_m}{(1 + \dot{A}_m \dot{F}_h)}$$

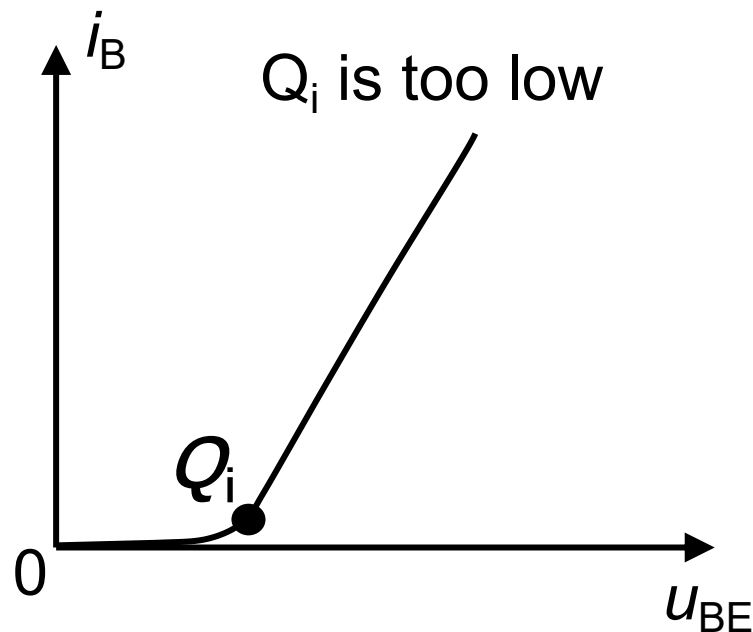
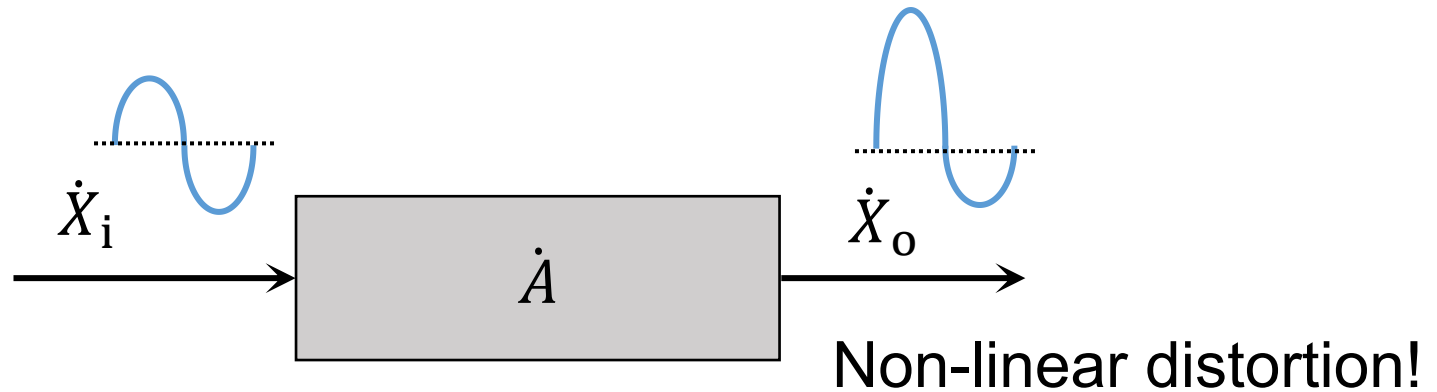
Bandwidth gain product 带宽增益积

$$|\dot{A}_{\text{mf}} f_{\text{bwf}}| = |\dot{A}_m f_{\text{bw}}|$$

The bandwidth increases at the cost of gain

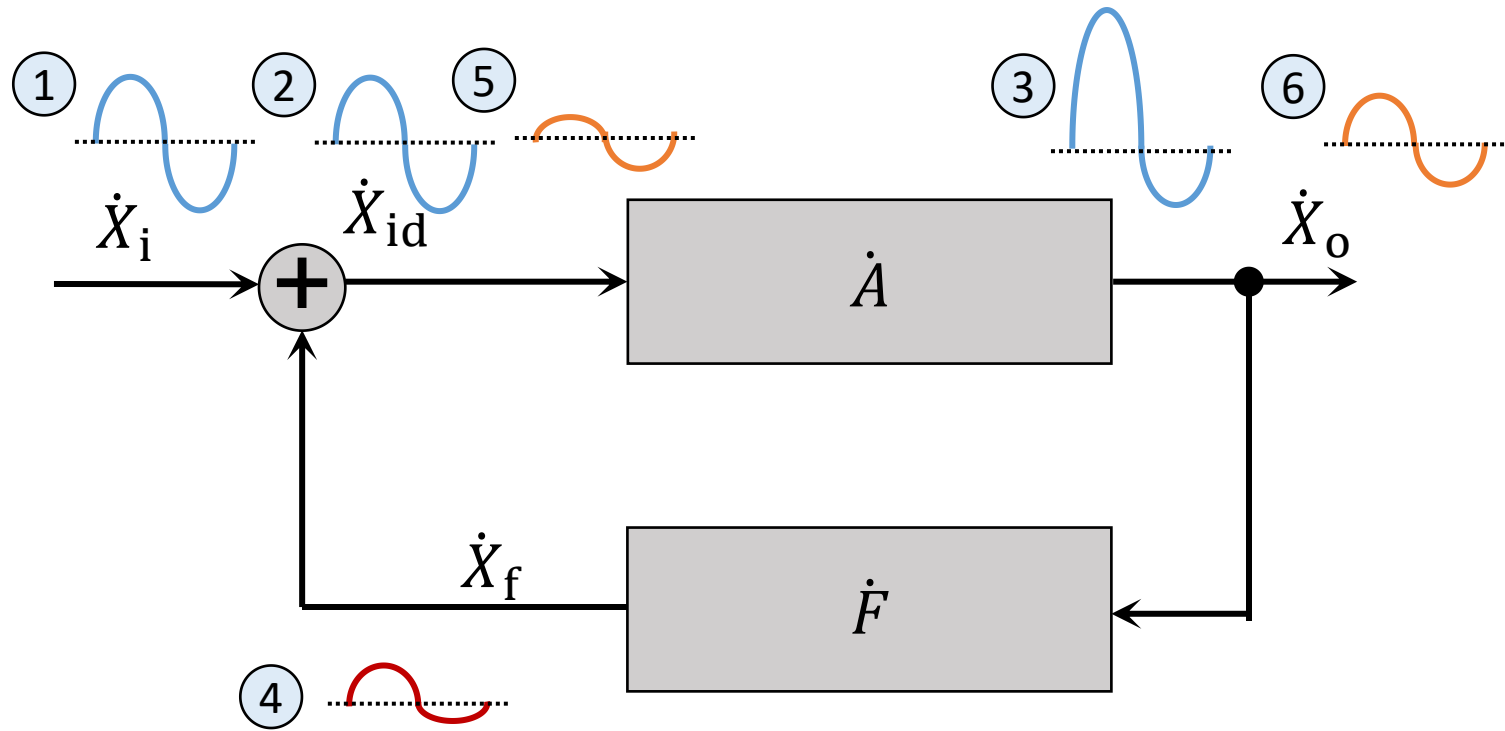
Negative feedback decreases the non-linear distortion

减小非线性失真



Negative feedback **decreases** the non-linear distortion

减小非线性失真



Effect of negative feedback on input resistance

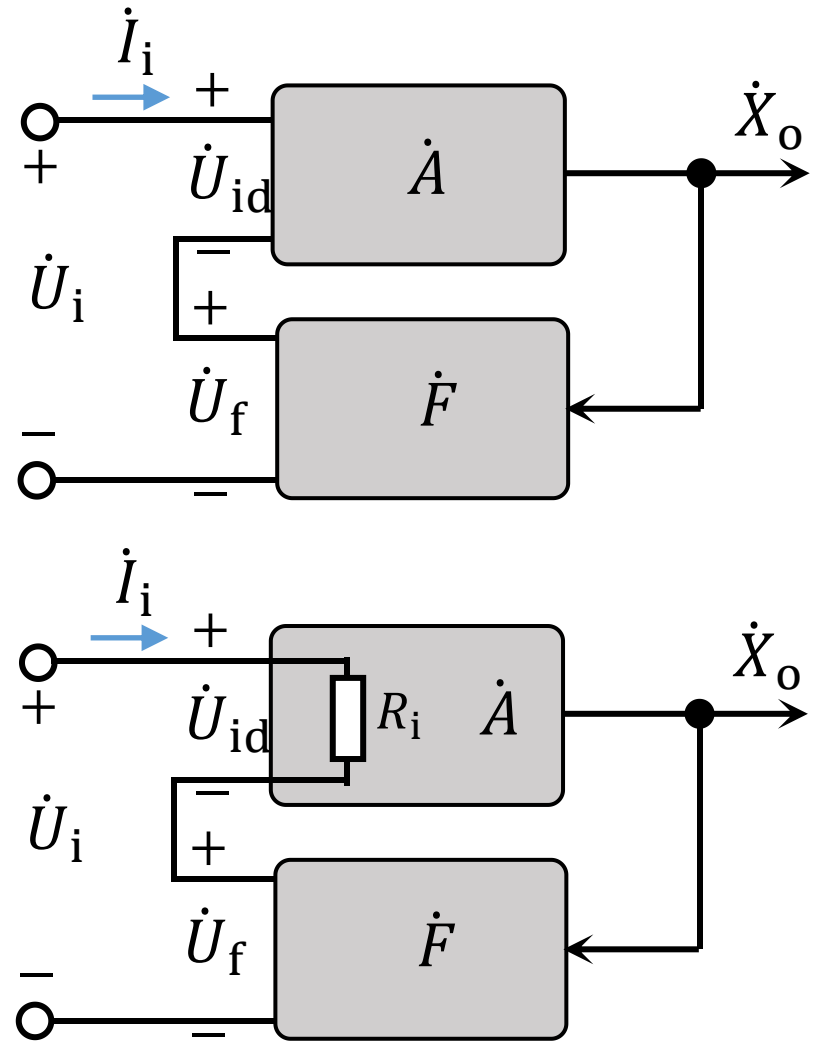
(1) Series negative feedback

$$R_i = \frac{\dot{U}_{id}}{\dot{I}_i}$$

$$R_{if} = \frac{\dot{U}_i}{\dot{I}_i} = \frac{\dot{U}_{id} + \dot{U}_f}{\dot{I}_i}$$
$$= \frac{\dot{U}_{id} + \dot{A}\dot{F}\dot{U}_{id}}{\dot{I}_i}$$

$$= (1 + \dot{A}\dot{F})R_i$$

Input resistance increase!



Effect of negative feedback on input resistance

(2) **Parallel** negative feedback

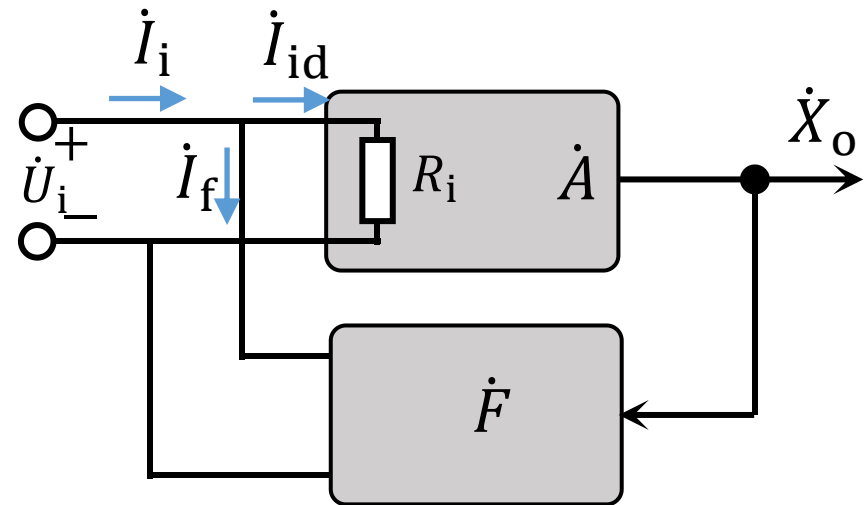
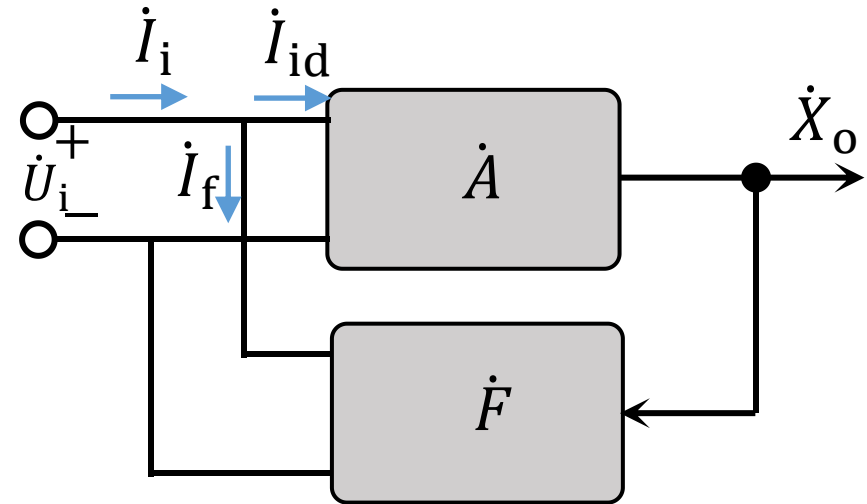
$$R_i = \frac{\dot{U}_i}{\dot{I}_{id}}$$

$$R_{if} = \frac{\dot{U}_i}{\dot{I}_i} = \frac{\dot{U}_i}{\dot{I}_{id} + \dot{I}_f}$$

$$= \frac{\dot{U}_i}{\dot{I}_{id} + \dot{A}\dot{F}\dot{I}_{id}}$$

$$= \frac{R_i}{1 + \dot{A}\dot{F}}$$

Input resistance **decrease!**



Effect of negative feedback on output resistance

No feedback network: R_o

(1) Voltage negative feedback

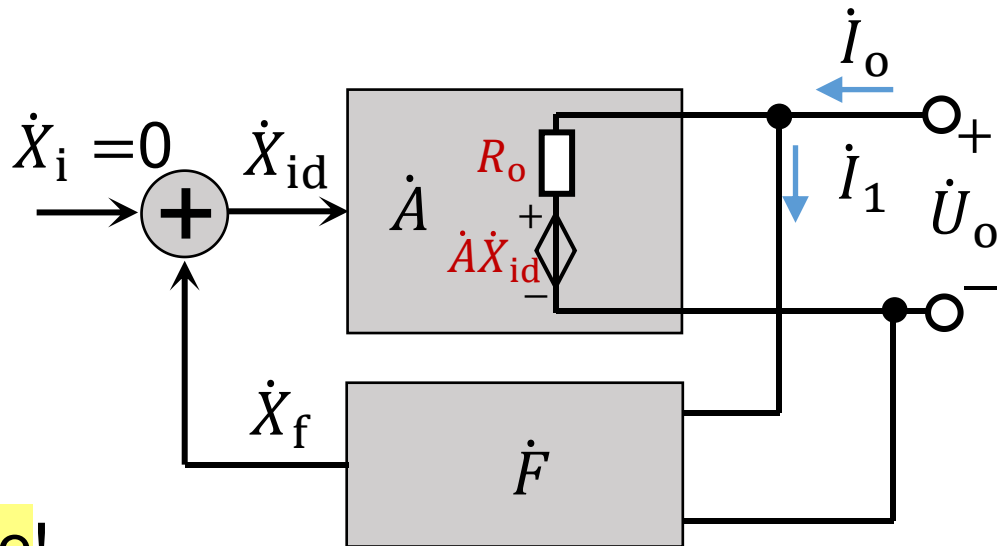
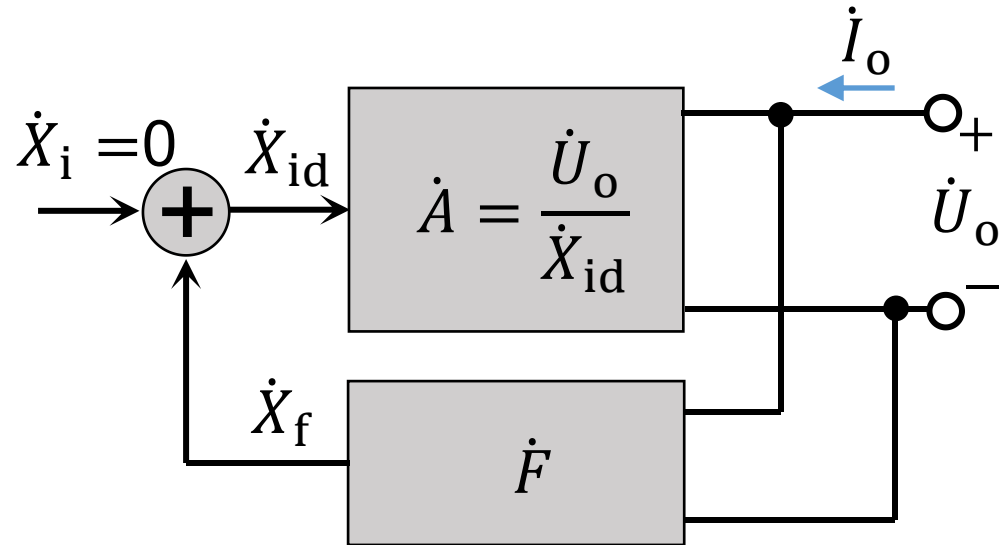
$$\begin{cases} R_{of} = \frac{\dot{U}_o}{\dot{I}_o} \\ R_o = \frac{\dot{U}_o - \dot{A}\dot{X}_{id}}{\dot{I}_o}, \text{ ignore } \dot{I}_1 \end{cases}$$

$$\dot{X}_f = \dot{F}\dot{U}_o$$

$$\dot{X}_{id} = -\dot{X}_f = -\dot{F}\dot{U}_o$$

$$R_{of} = \frac{R_o}{1 + \dot{A}\dot{F}}$$

Output resistance decrease!



Effect of negative feedback on output resistance

(2) Current negative feedback

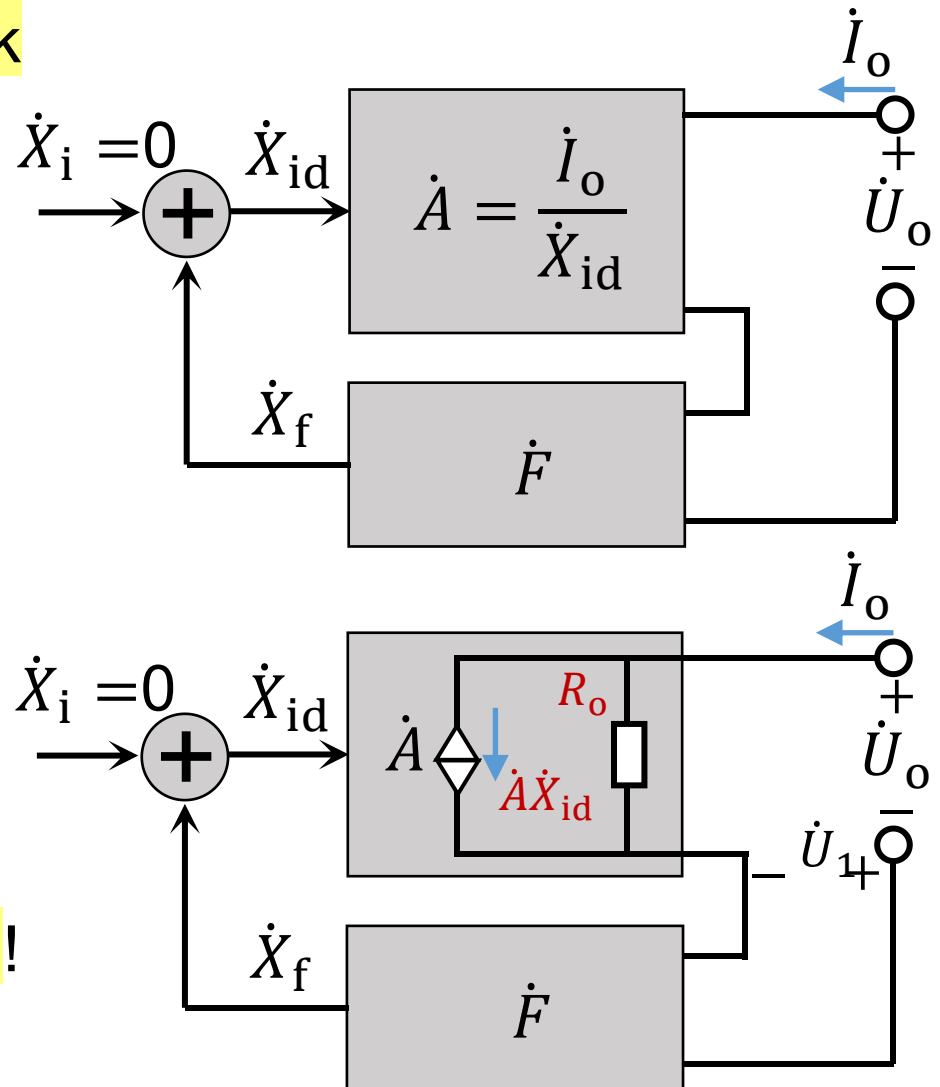
$$\left\{ \begin{array}{l} R_{of} = \frac{\dot{U}_o}{\dot{I}_o} \\ R_o = \frac{\dot{U}_o}{\dot{I}_o - \dot{A}\dot{X}_{id}} \text{ , ignore } \dot{U}_1 \end{array} \right.$$

$$\dot{X}_f = \dot{F}\dot{I}_o$$

$$\dot{X}_{id} = -\dot{X}_f = -\dot{F}\dot{I}_o$$

$$R_{of} = (1 + \dot{A}\dot{F})R_o$$

Output resistance **increase!**



Summary

Series negative feedback

Input $R_i \uparrow$

Parallel negative feedback

Input $R_i \downarrow$

Voltage negative feedback

Output $R_o \downarrow$

Stabilize the output U_o

Constant-voltage source
恒压源

Current negative feedback

Output $R_o \uparrow$

Stabilize the output I_o

Constant-current source
恒流源

How to choose the right negative feedback type to design electronic circuits? 如何正确选择负反馈类型进行电路设计?

For voltage amplifier:

Choose voltage series negative feedback

For current amplifier:

Choose current parallel negative feedback

For voltage-current converter:

Choose current series negative feedback

For current-voltage converter:

Choose voltage parallel negative feedback

How to choose the right negative feedback type to design electronic circuits? 如何正确选择负反馈类型进行电路设计?

Choose proper amount of feedback: $1 + \dot{A}\dot{F}$

选择合适的反馈深度

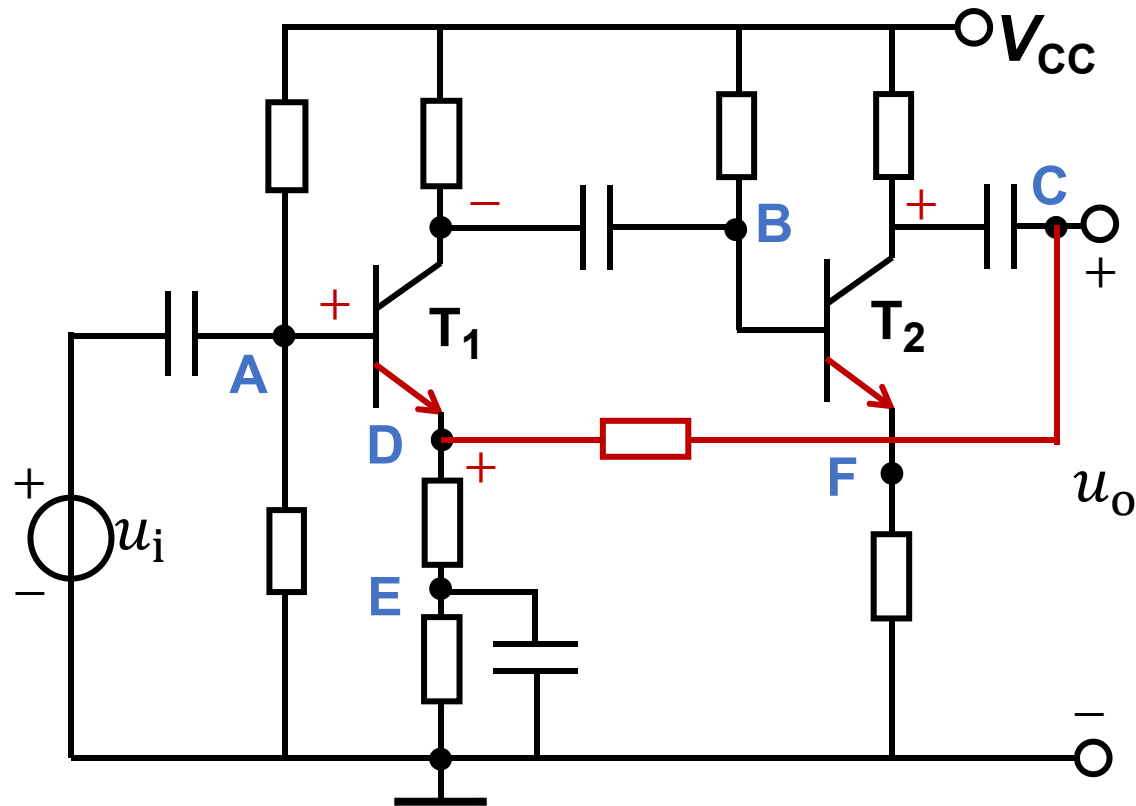
1. Use integrated operational amplifier 采用集成运算放大器

$\dot{A}$, R_i , R_o are fixed.

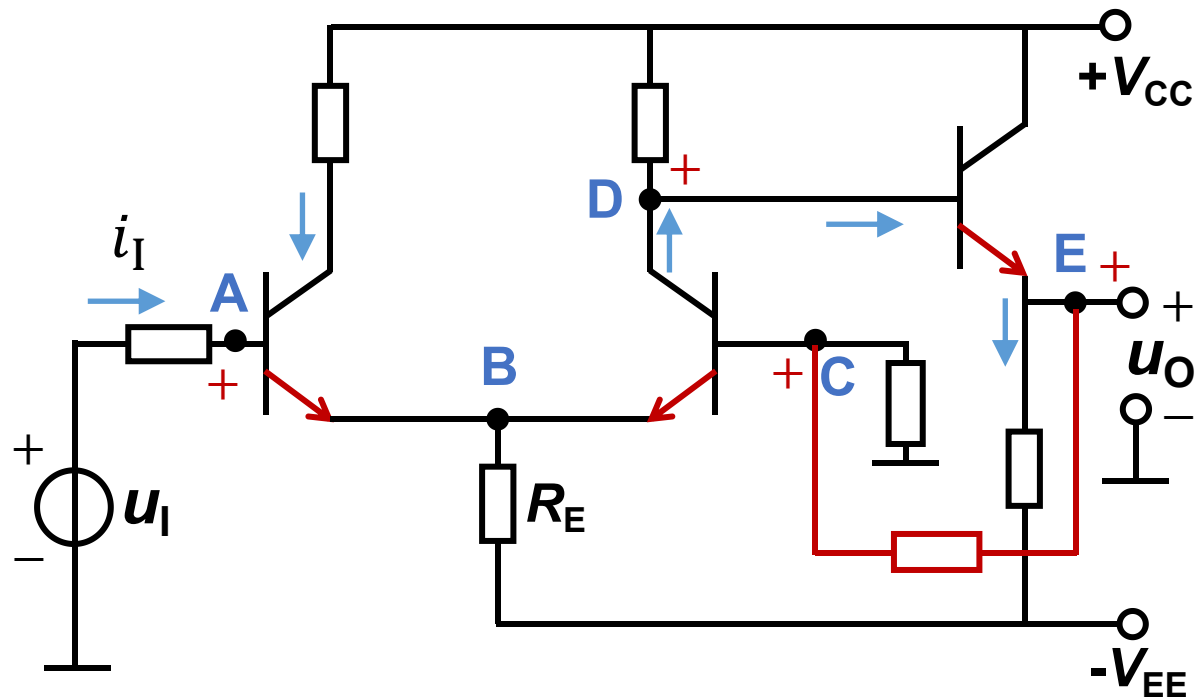
2. Estimate the feedback factor $\dot{F}$

3. Design feedback network

Example 1: To stabilize the AC output voltage u_o , the feedback resistor is put between ? and ? Points. 



Example 2: To stabilize the AC output voltage u_o , the feedback resistor is put between X and X Points. 



6.4 Amplifier circuit with strong negative feedback

Close-loop gain 闭环增益

$$\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$

For negative feedback, $\dot{A}\dot{F} > 0$: $\dot{A}_f < \dot{A}$

When $\dot{A}\dot{F} \gg 1$:

$$\dot{A}_f \approx \frac{1}{\dot{F}}$$

This is called **strong negative feedback** 深度负反馈.

- ❑ The close-loop gain only depends on the feedback network.
- ❑ Since feedback network is usually made of resistors and capacitors, the amplifier circuit is very stable.

For strong negative feedback:

$$\dot{A}_f \approx \frac{1}{\dot{F}}$$

$$\left\{ \begin{array}{l} \dot{A}_f = \frac{\dot{X}_o}{\dot{X}_i} \\ \dot{F} = \frac{\dot{X}_f}{\dot{X}_o} \end{array} \right. \Rightarrow \dot{X}_i \approx \dot{X}_f \quad \dot{X}_{id} \approx 0$$

Virtual short circuit

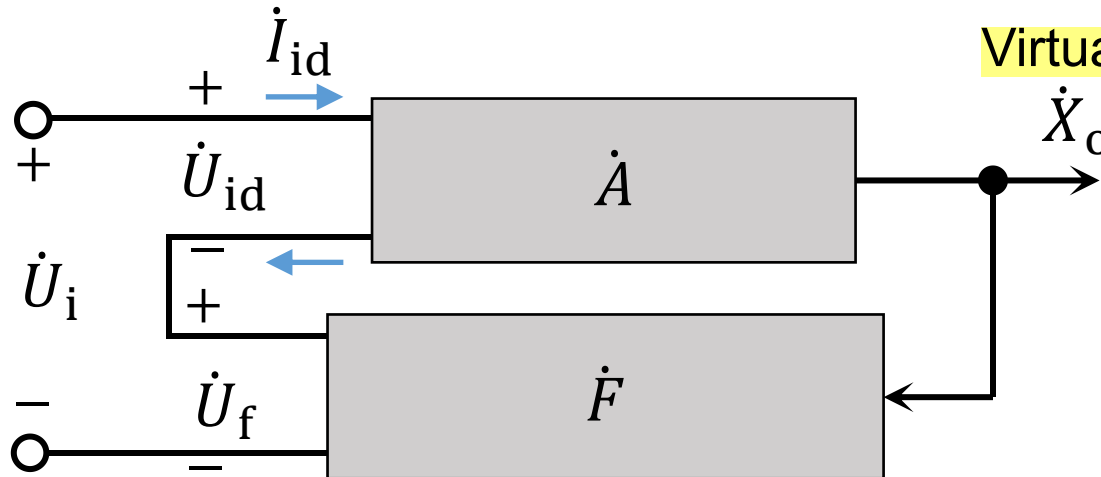
For series feedback:

$$\dot{U}_i \approx \dot{U}_f$$

$$\dot{U}_{id} \approx 0$$

$$\dot{I}_{id} \approx 0$$

Virtual open circuit



For strong negative feedback:

$$\dot{A}_f \approx \frac{1}{\dot{F}}$$

$$\left\{ \begin{array}{l} \dot{A}_f = \frac{\dot{X}_o}{\dot{X}_i} \\ \dot{F} = \frac{\dot{X}_f}{\dot{X}_o} \end{array} \right. \Rightarrow \dot{X}_i \approx \dot{X}_f \quad \dot{X}_{id} \approx 0$$

Virtual open circuit

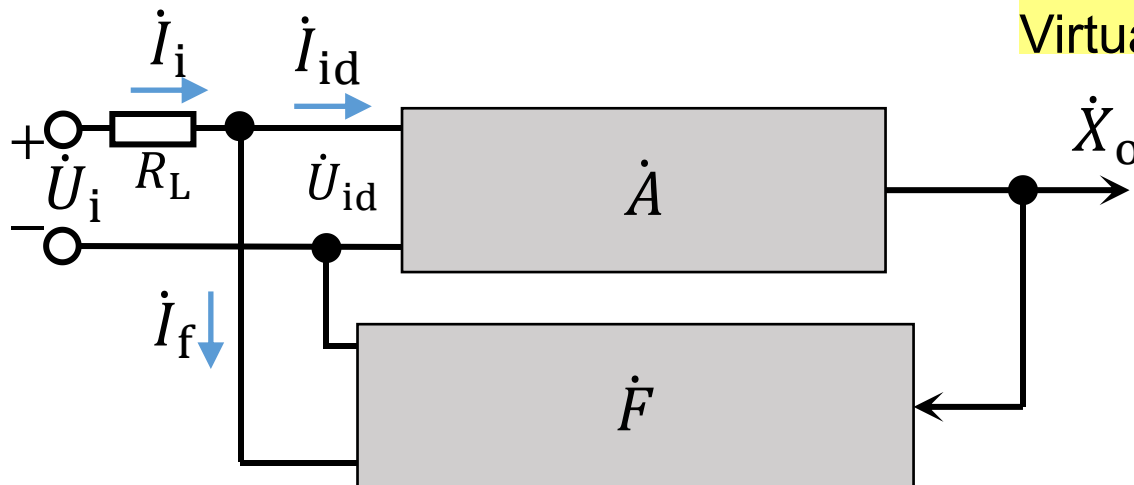
For parallel feedback:

$$\dot{I}_i \approx \dot{I}_f$$

$$\dot{I}_{id} \approx 0$$

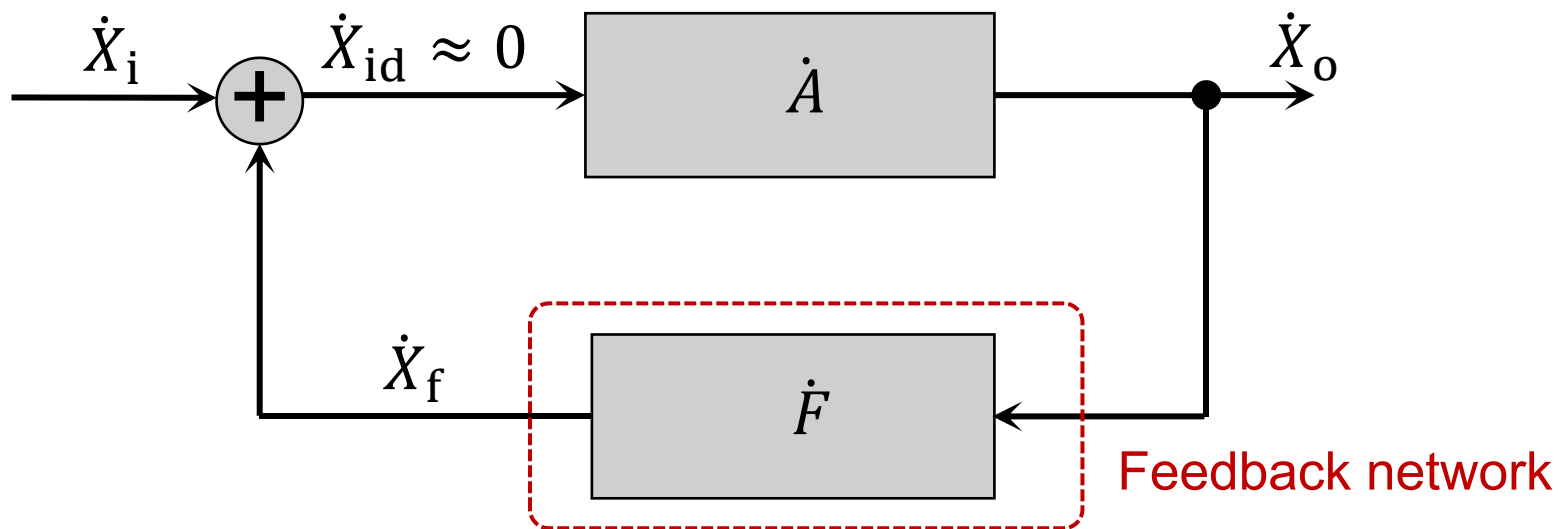
$$\dot{U}_{id} \approx 0$$

Virtual short circuit

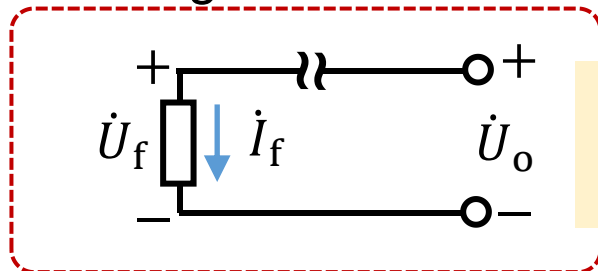


How to get $\dot{F}$ and $\dot{A}_f \approx \frac{1}{\dot{F}}$?

Separate amplifier circuit and feedback network
分离放大电路和反馈网络

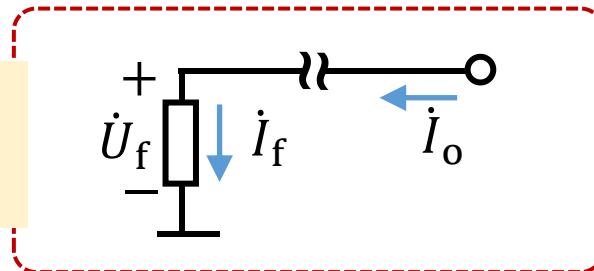


Voltage feedback



$$\dot{F} = \frac{\dot{X}_f}{\dot{X}_o}$$

Current feedback



Example 1: Voltage series negative feedback 电压串联负反馈

Close-loop gain

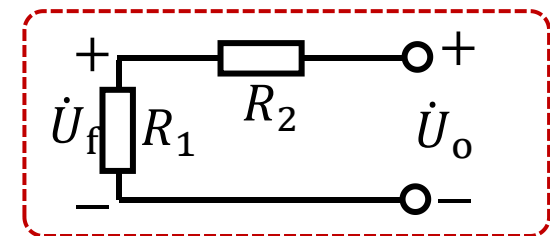
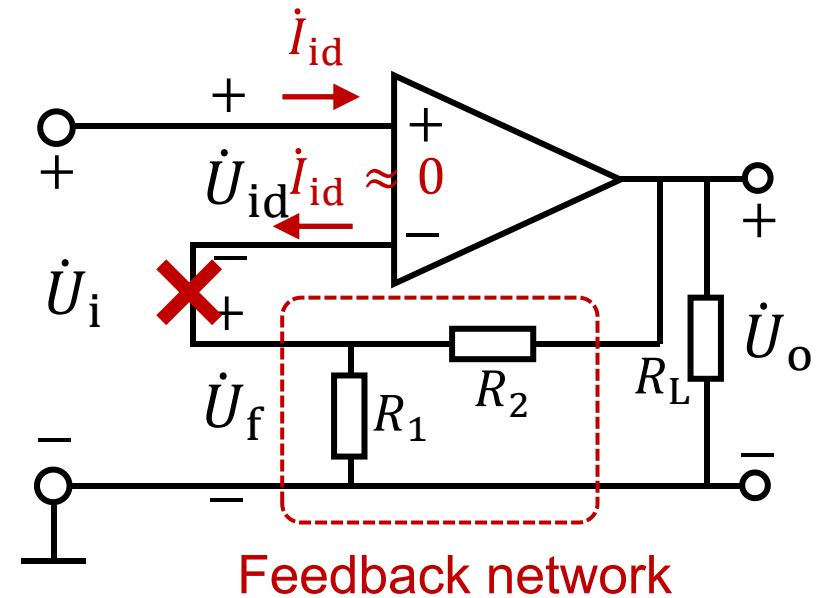
$$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_i} \approx \frac{\dot{U}_o}{\dot{U}_f} = \frac{1}{\dot{F}_u}$$

$$\dot{U}_f = \frac{R_1}{R_1 + R_2} \dot{U}_o$$

$$\dot{F}_u = \frac{R_1}{R_1 + R_2}$$

$$\dot{A}_{uf} \approx \frac{\dot{U}_o}{\dot{U}_f} = \frac{R_1 + R_2}{R_1}$$

Independent of load resistor!



Non-inverted input scaling
amplifier

同相输入比例放大器

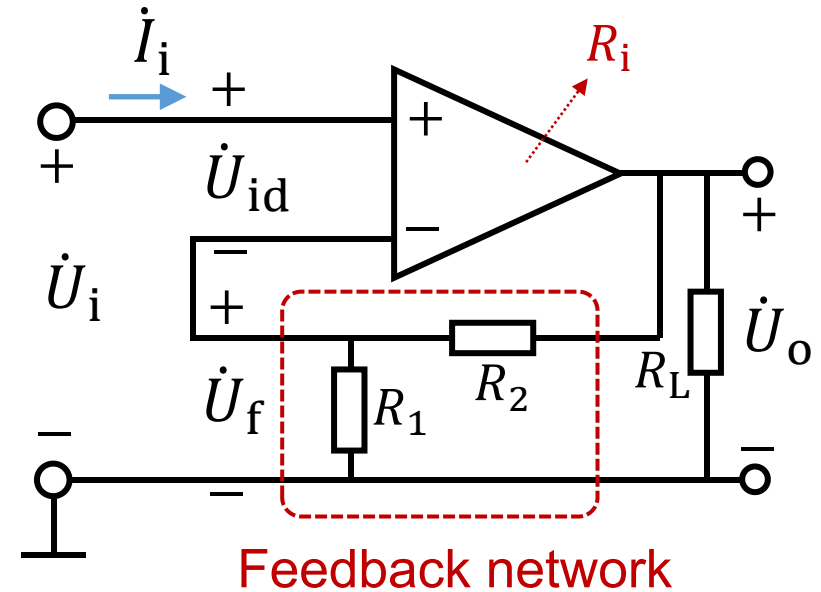
Example 1: Voltage series negative feedback 电压串联负反馈

Input resistance:

$$R_i = \frac{\dot{U}_{id}}{\dot{I}_i}$$

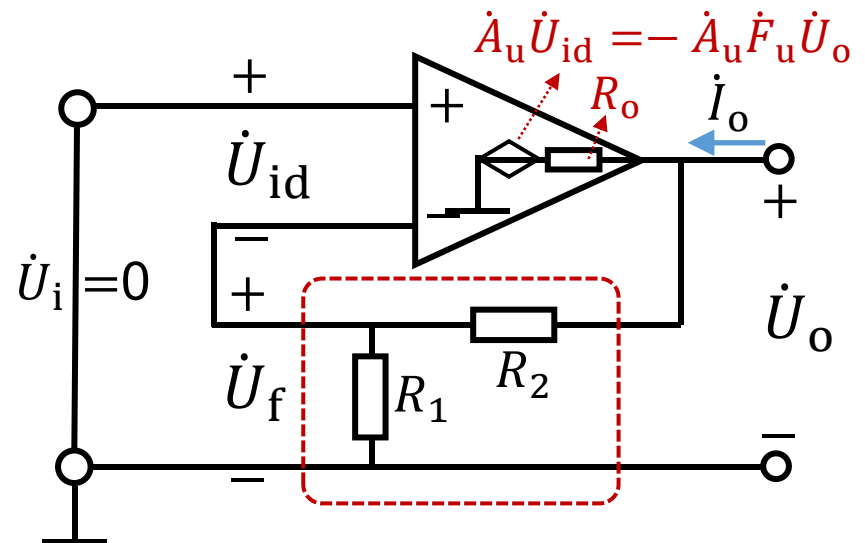
$$R_{if} = \frac{\dot{U}_i}{\dot{I}_i} = \frac{\dot{U}_{id} + \dot{U}_f}{\dot{I}_i}$$

$$= (1 + \dot{A}_u \dot{F}_u) R_i \rightarrow \infty$$



Output resistance:

$$R_{of} = \frac{R_o}{1 + \dot{A}_u \dot{F}_u} \approx 0$$



Example 2: Voltage parallel negative feedback 电压并联负反馈

Close-loop gain

$$\dot{A}_{\text{rf}} = \frac{\dot{U}_o}{\dot{I}_i} \approx \frac{\dot{U}_o}{\dot{I}_f} = \frac{1}{\dot{F}_g}$$

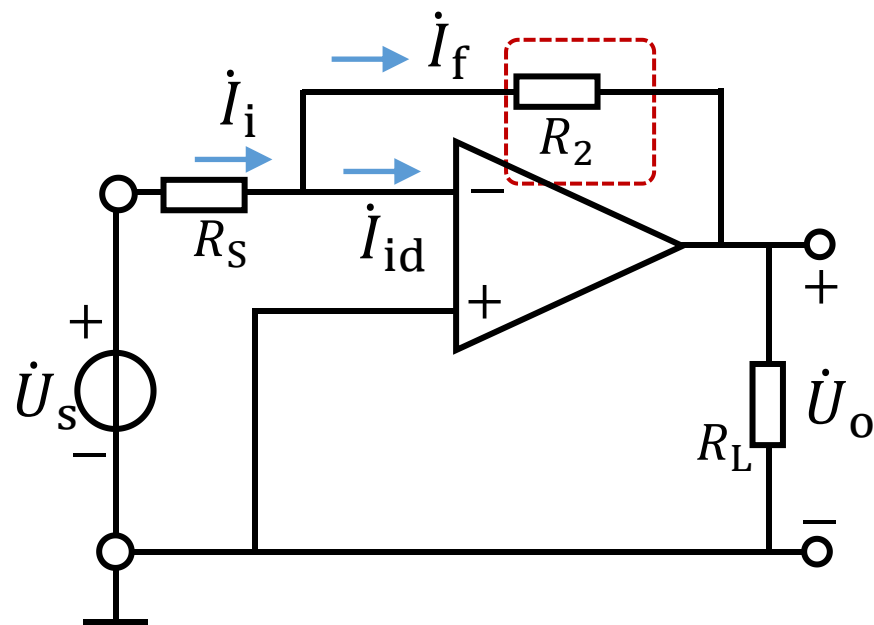
$$\dot{I}_f = -\frac{\dot{U}_o}{R_2}$$

$$\dot{F}_g = -\frac{1}{R_2}$$

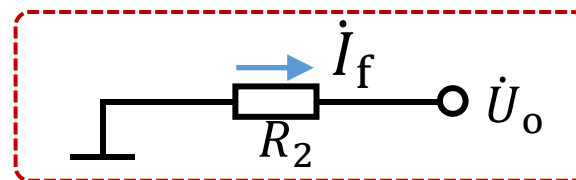
$$\dot{A}_{\text{rf}} \approx \frac{\dot{U}_o}{\dot{I}_f} = -R_2$$

Voltage gain

$$\dot{A}_{\text{usf}} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{U}_o}{\dot{I}_i R_s} = -\frac{R_2}{R_s}$$



Feedback network



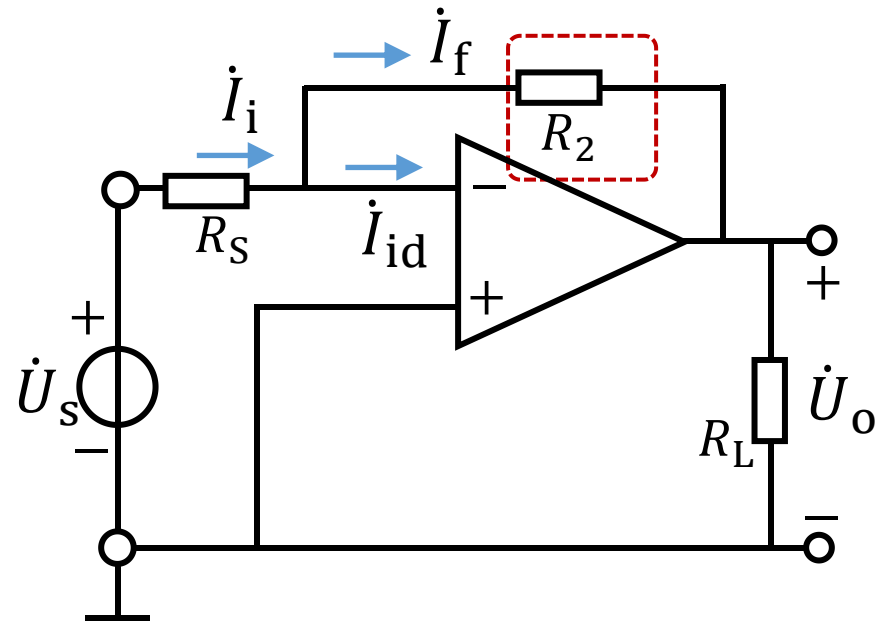
Example 2: Voltage parallel negative feedback 电压并联负反馈

Input resistance:

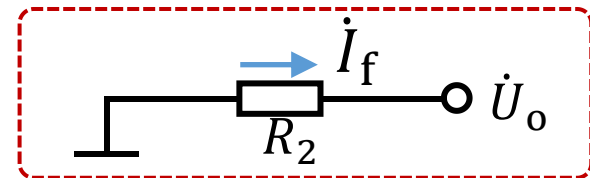
$$R_i = \frac{\dot{U}_{id}}{\dot{I}_{id}}$$
$$R_{if} = R_s + \frac{\dot{U}_{id}}{\dot{I}_i} = R_s + \frac{\dot{U}_{id}}{\dot{I}_f + \dot{I}_{id}}$$
$$= R_s + \frac{R_i}{1 + \dot{A}_r \dot{F}_g} \approx R_s$$

Output resistance:

$$R_{of} = \frac{R_o}{1 + \dot{A}_r \dot{F}_g} \approx 0$$



Feedback network



Example 3: Current series negative feedback 电流串联负反馈

Close-loop gain

$$\dot{A}_{\text{gf}} = \frac{\dot{I}_o}{\dot{U}_i} \approx \frac{\dot{I}_o}{\dot{U}_f} = \frac{1}{\dot{F}_r}$$

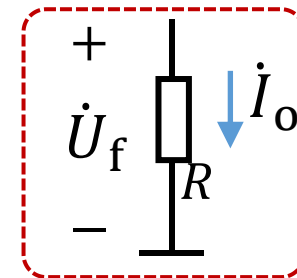
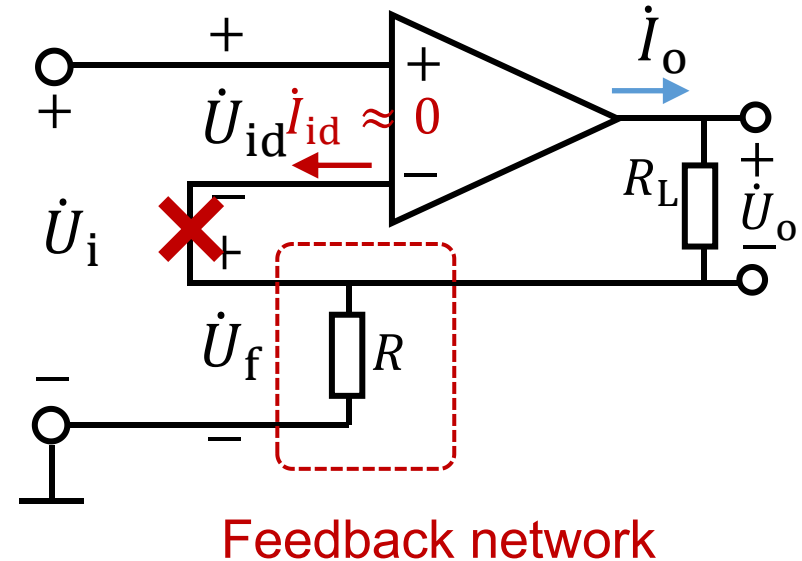
$$\dot{U}_f = \dot{I}_o R$$

$$\dot{F}_r = R$$

$$\dot{A}_{\text{gf}} \approx \frac{\dot{I}_o}{\dot{U}_f} = \frac{1}{R}$$

Voltage gain

$$\dot{A}_{\text{uf}} = \frac{\dot{U}_o}{\dot{U}_i} \approx \frac{\dot{U}_o}{\dot{U}_f} = \frac{\dot{I}_o R_L}{\dot{I}_o R} = \frac{R_L}{R}$$



Example 3: Current series negative feedback 电流串联负反馈

Input resistance:

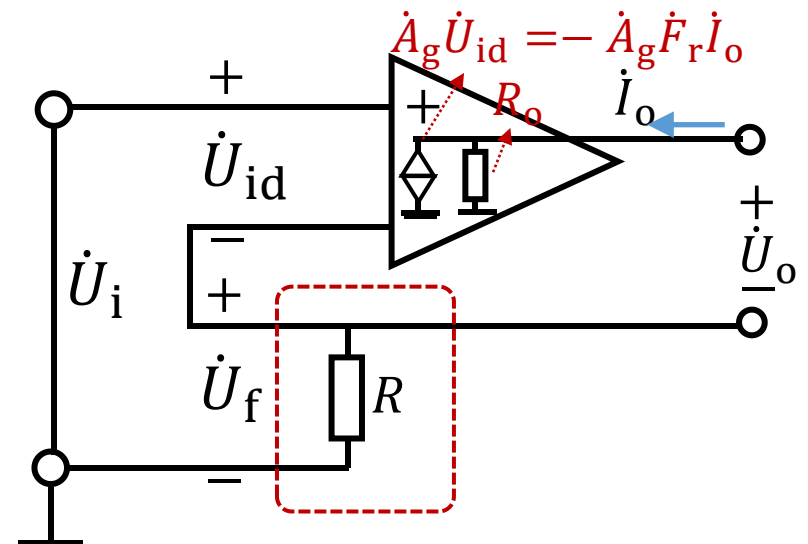
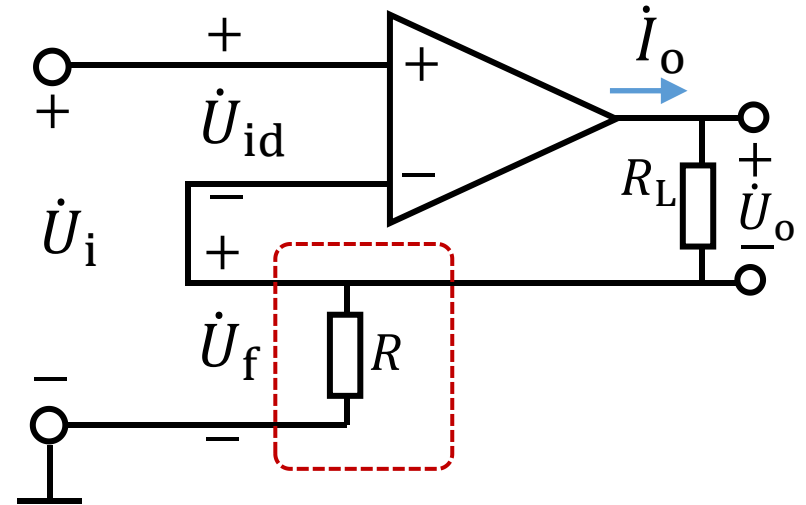
$$R_{if} = \frac{\dot{U}_i}{\dot{I}_i} = \frac{\dot{U}_{id} + \dot{U}_f}{\dot{I}_i}$$

$$= (1 + \dot{A}_g \dot{F}_r) R_i \rightarrow \infty$$

Output resistance:

$$\dot{I}_o = \frac{\dot{U}_o}{R_o} + (-\dot{A}_g \dot{F}_r \dot{I}_o)$$

$$R_{of} = R_o (1 + \dot{A}_g \dot{F}_r) \rightarrow \infty$$



Example 4: Current parallel negative feedback 电流并联负反馈

Close-loop gain

$$\dot{A}_{\text{if}} = \frac{\dot{I}_o}{\dot{I}_i} \approx \frac{\dot{I}_o}{\dot{I}_f} = \frac{1}{\dot{F}_i}$$

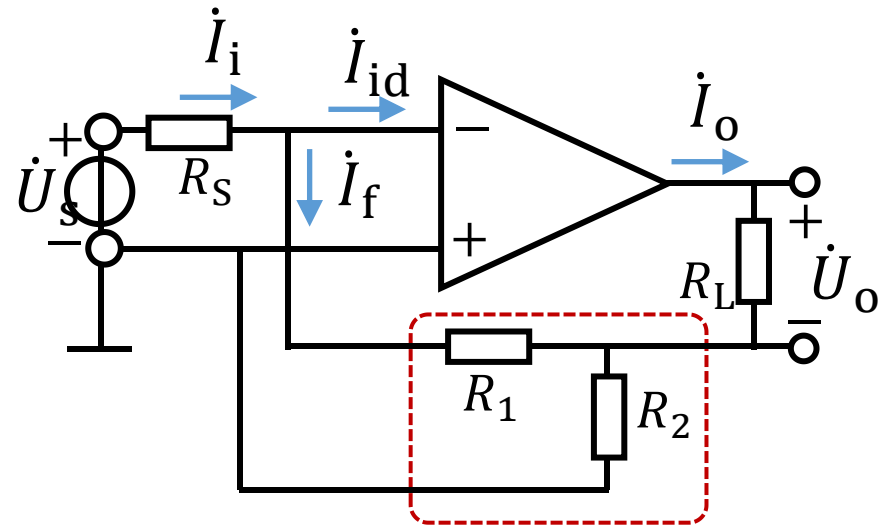
$$\dot{I}_f = -\dot{I}_o \frac{R_2}{R_1 + R_2}$$

$$\dot{F}_i = -\frac{R_2}{R_1 + R_2}$$

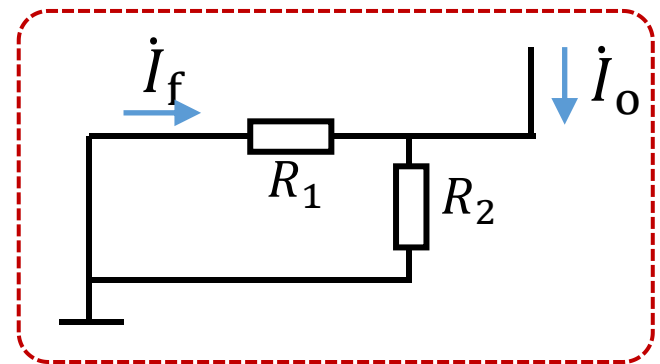
$$\dot{A}_{\text{if}} \approx \frac{\dot{I}_o}{\dot{I}_f} = -\frac{R_1 + R_2}{R_2}$$

Voltage gain

$$\dot{A}_{\text{usf}} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{I}_o R_L}{\dot{I}_i R_s} = -\frac{R_1 + R_2}{R_2} \frac{R_L}{R_s}$$



Feedback network



Example 4: Current parallel negative feedback 电流并联负反馈

Input resistance:

$$R_i = \frac{\dot{U}_{id}}{\dot{I}_{id}}$$

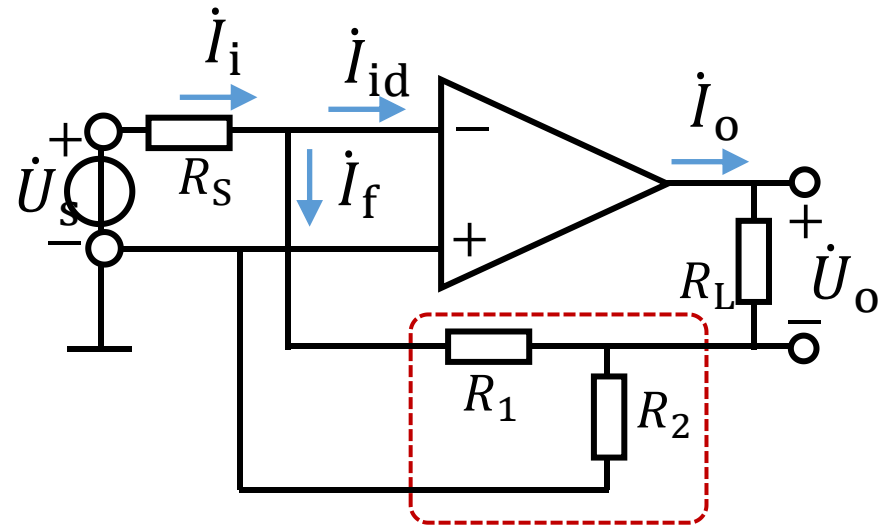
$$R_{if} = R_s + \frac{\dot{U}_{id}}{\dot{I}_i} = R_s + \frac{\dot{U}_{id}}{\dot{I}_f + \dot{I}_{id}}$$

$$= R_s + \frac{R_i}{1 + \dot{A}_r \dot{F}_g} \approx R_s$$

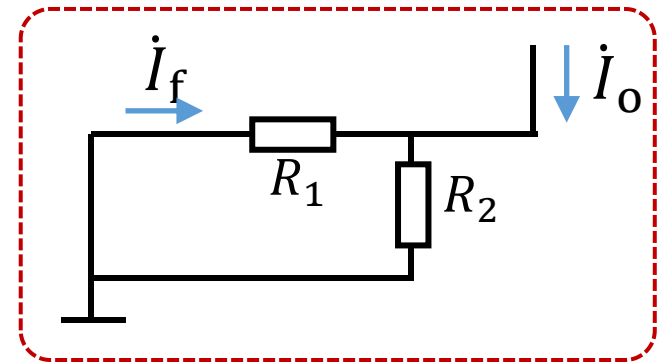
Output resistance:

$$\dot{I}_o = \frac{\dot{U}_o}{R_o} + (-\dot{A}_g \dot{F}_r \dot{I}_o)$$

$$R_{of} = R_o(1 + \dot{A}_g \dot{F}_r) \rightarrow \infty$$

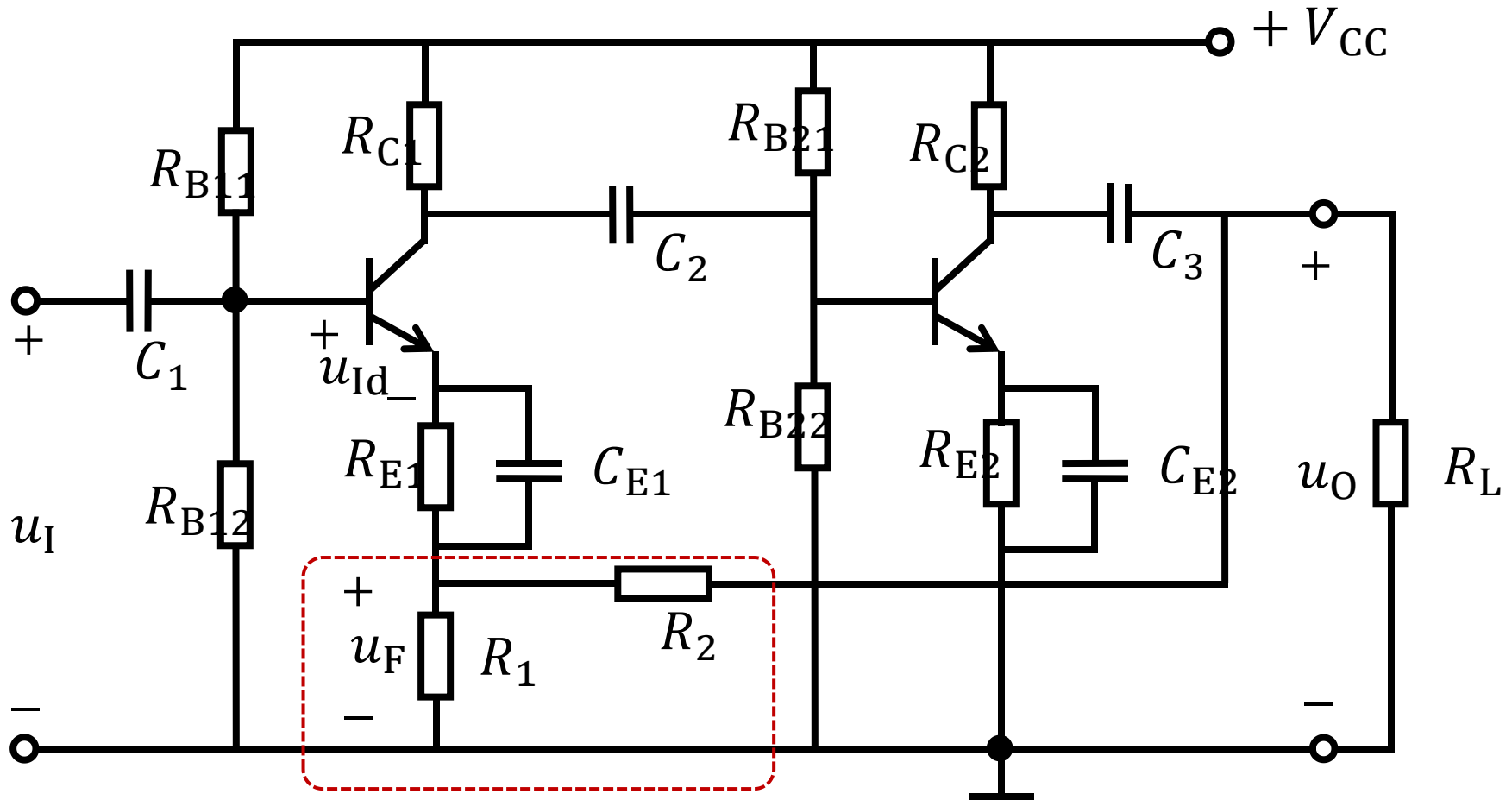


Feedback network



Example 5: strong negative feedback

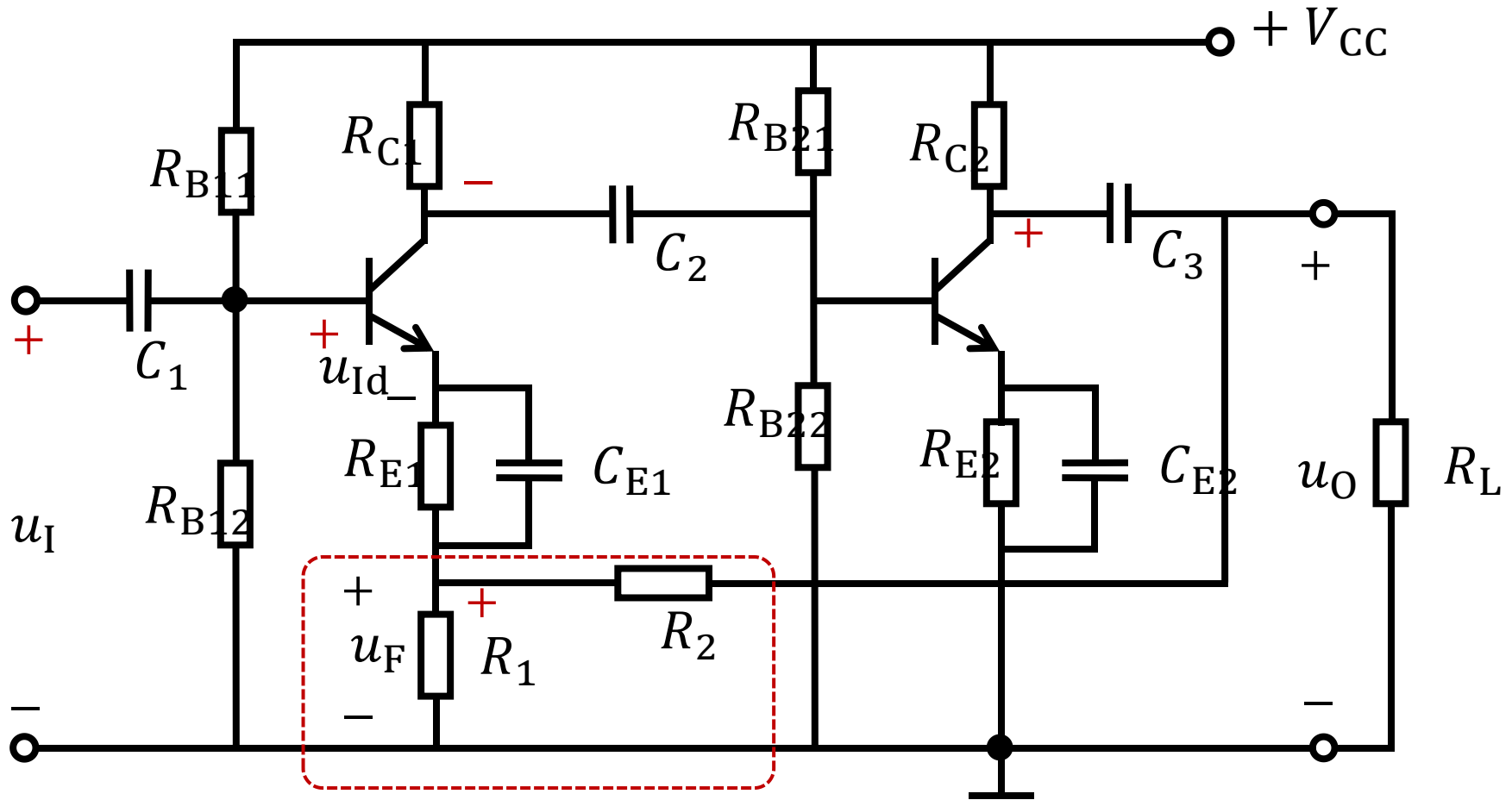
(1) Find the feedback network for AC circuit



Feedback network

Example 5:

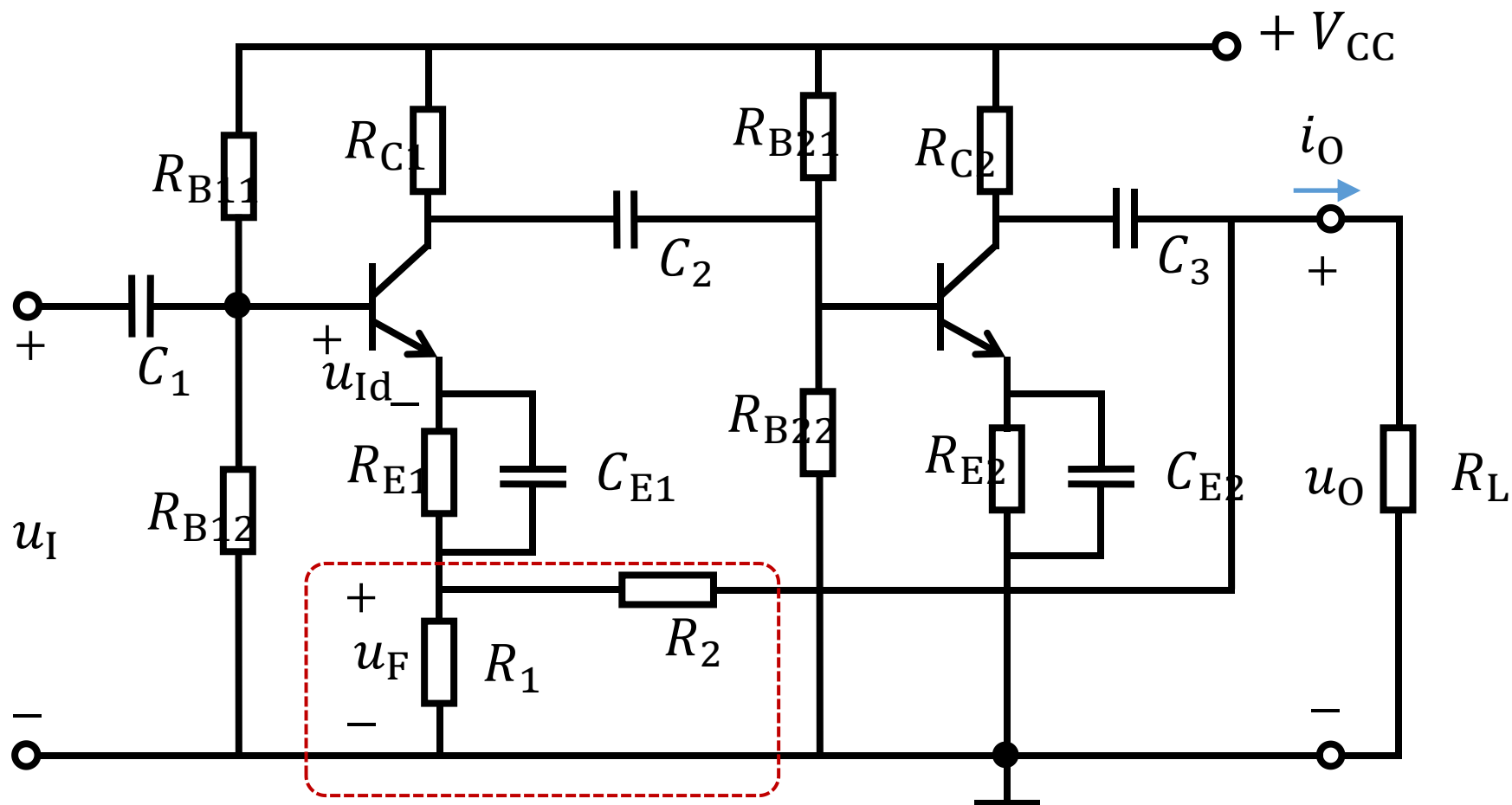
(2) Negative or positive AC feedback circuit



Negative feedback!

Example 5:

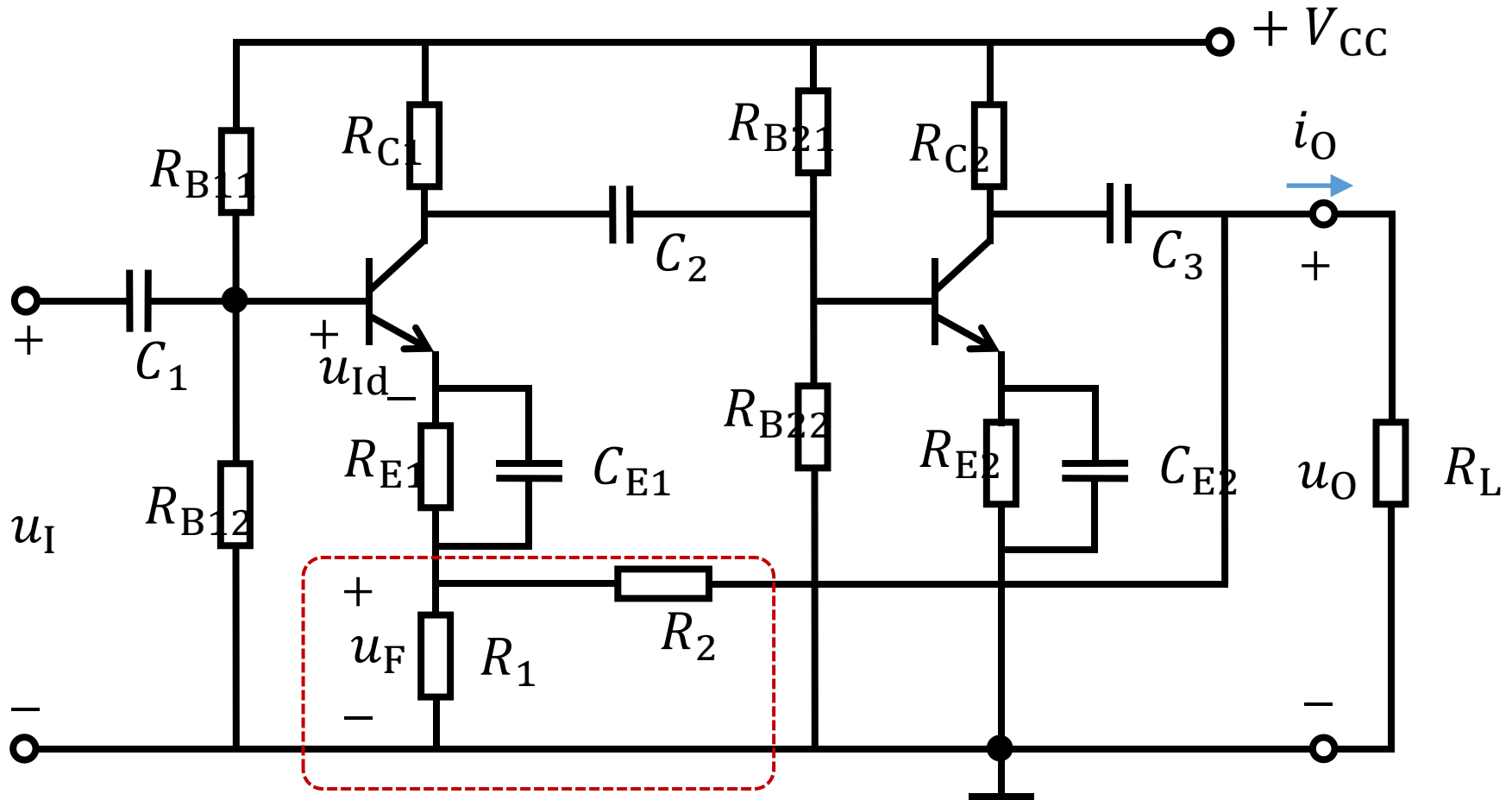
(3) Judge the type of AC negative feedback circuit



Voltage series feedback 电压串联负反馈

Example 5:

(4) Feedback factor and close-loop gain

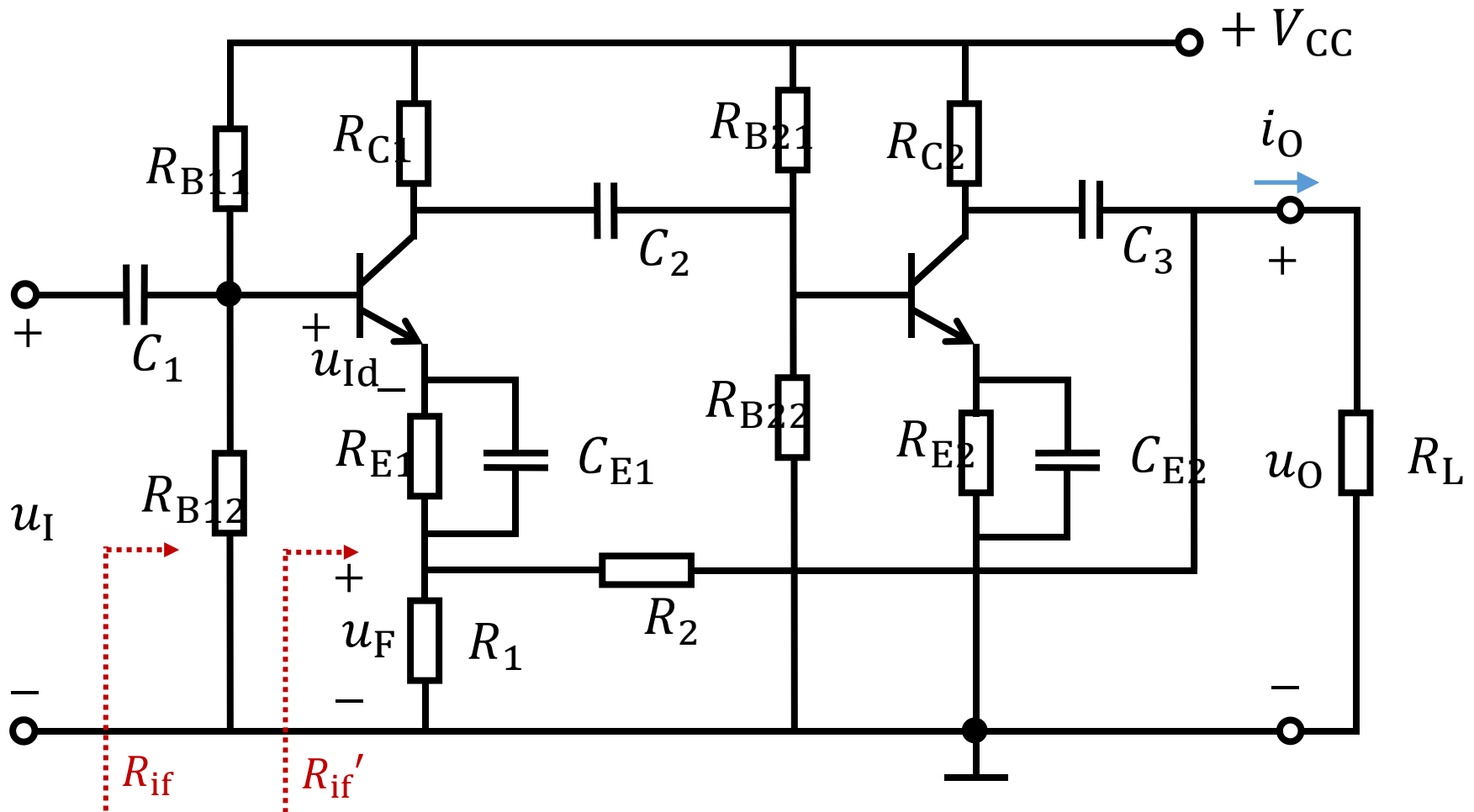


$$\dot{F}_u = \frac{\dot{U}_f}{\dot{U}_o} = \frac{R_1}{R_1 + R_2}$$

$$\dot{A}_{uf} \approx \frac{1}{\dot{F}_u} = \frac{R_1 + R_2}{R_1}$$

Example 5:

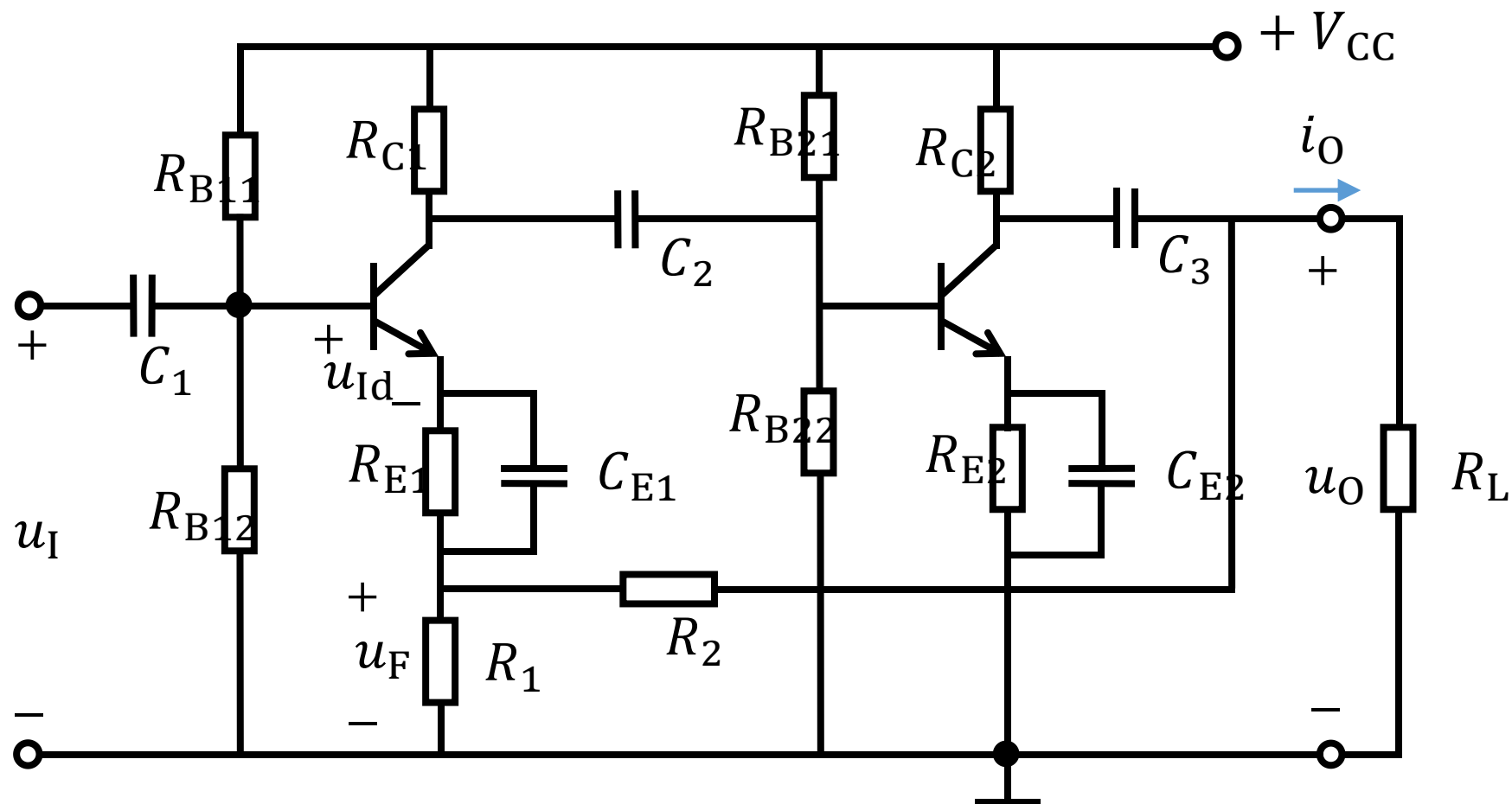
(5) Input resistance



$$R_{if} = R_{B12} // R_{B11} // R_{if}' \approx R_{B12} // R_{B11}$$

Example 5:

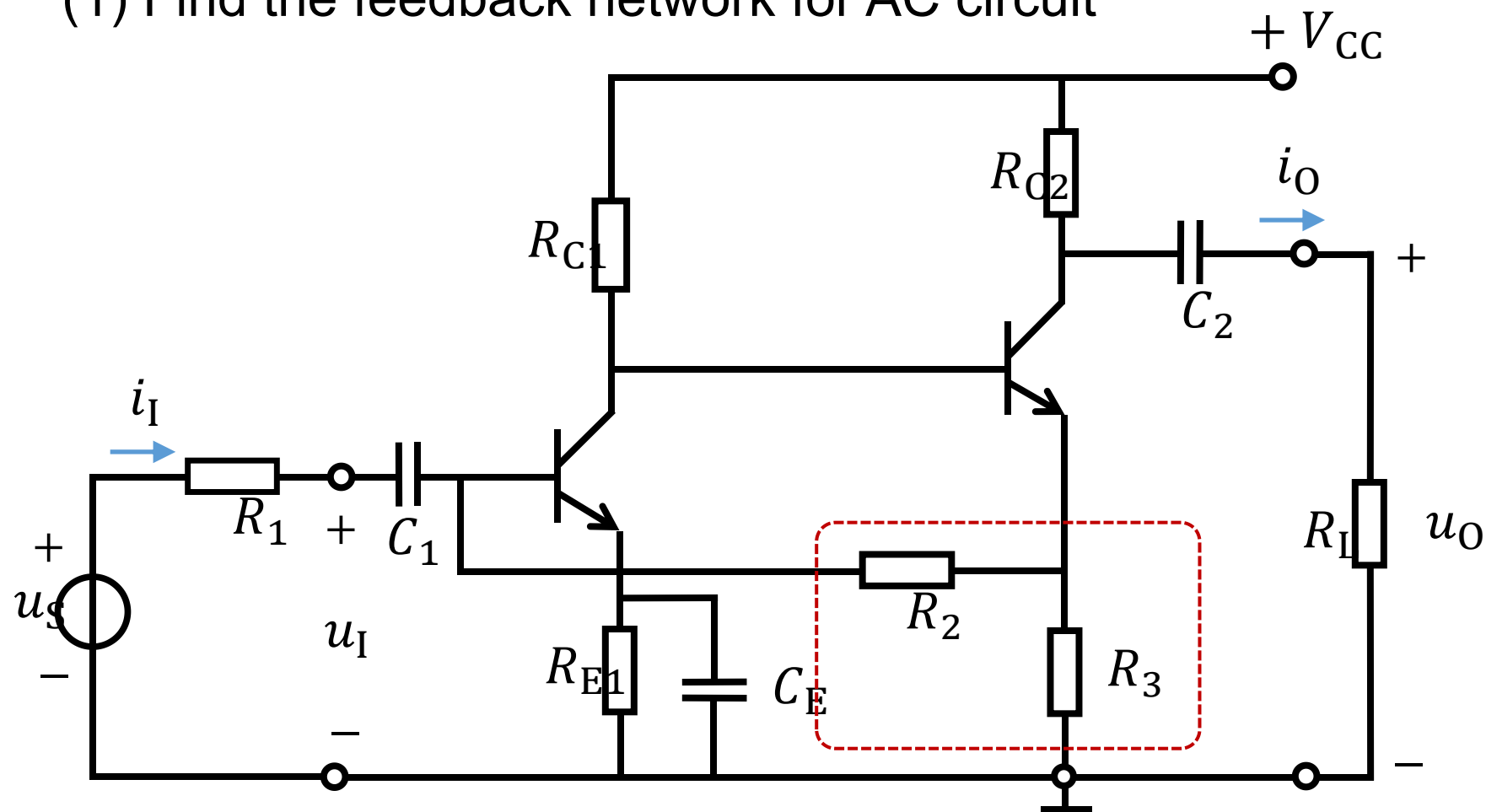
(6) Output resistance



$$R_o \approx 0$$

Example 6:

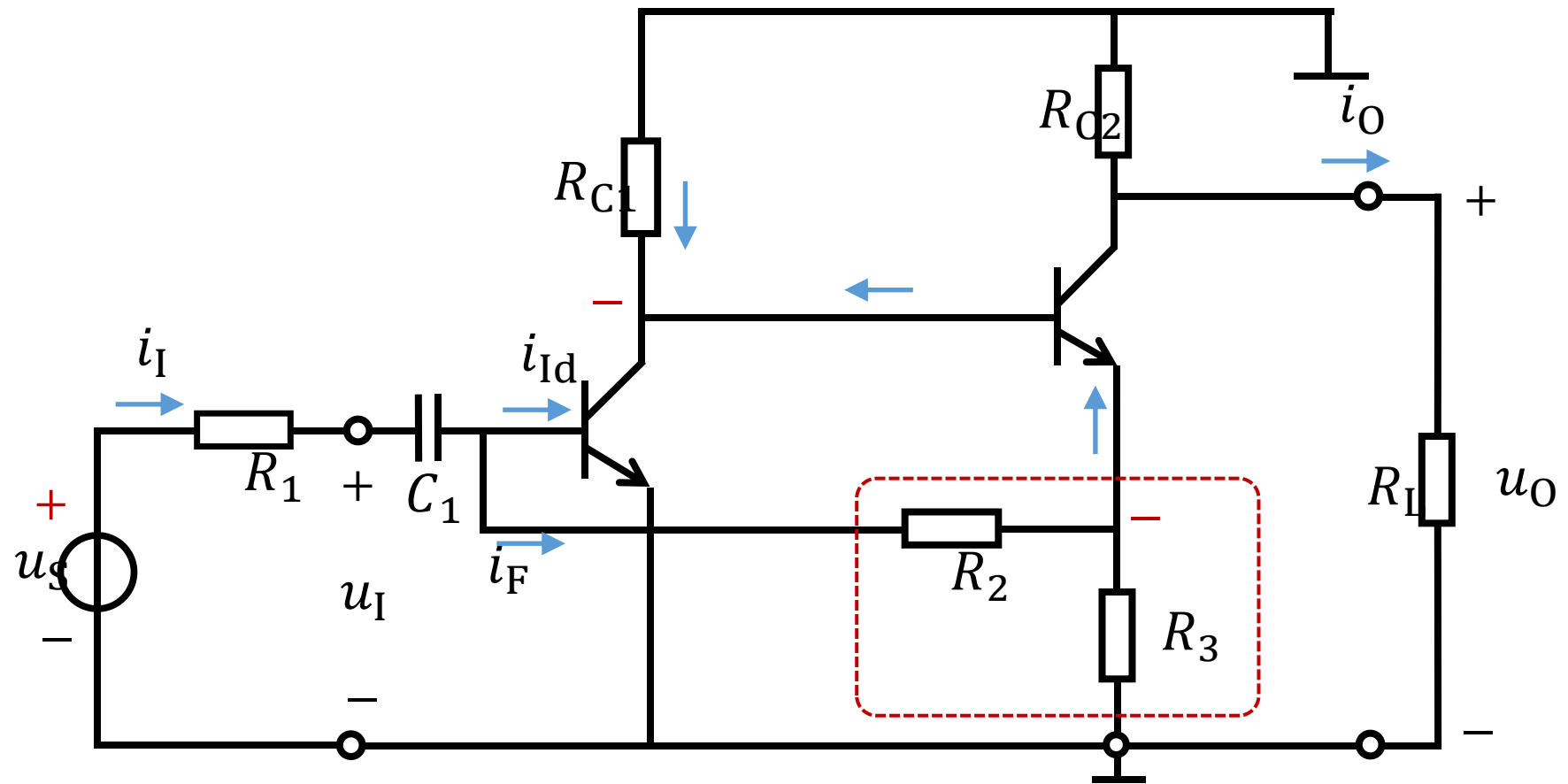
(1) Find the feedback network for AC circuit



Feedback network

Example 6:

(2) Positive or negative feedback for AC circuit?

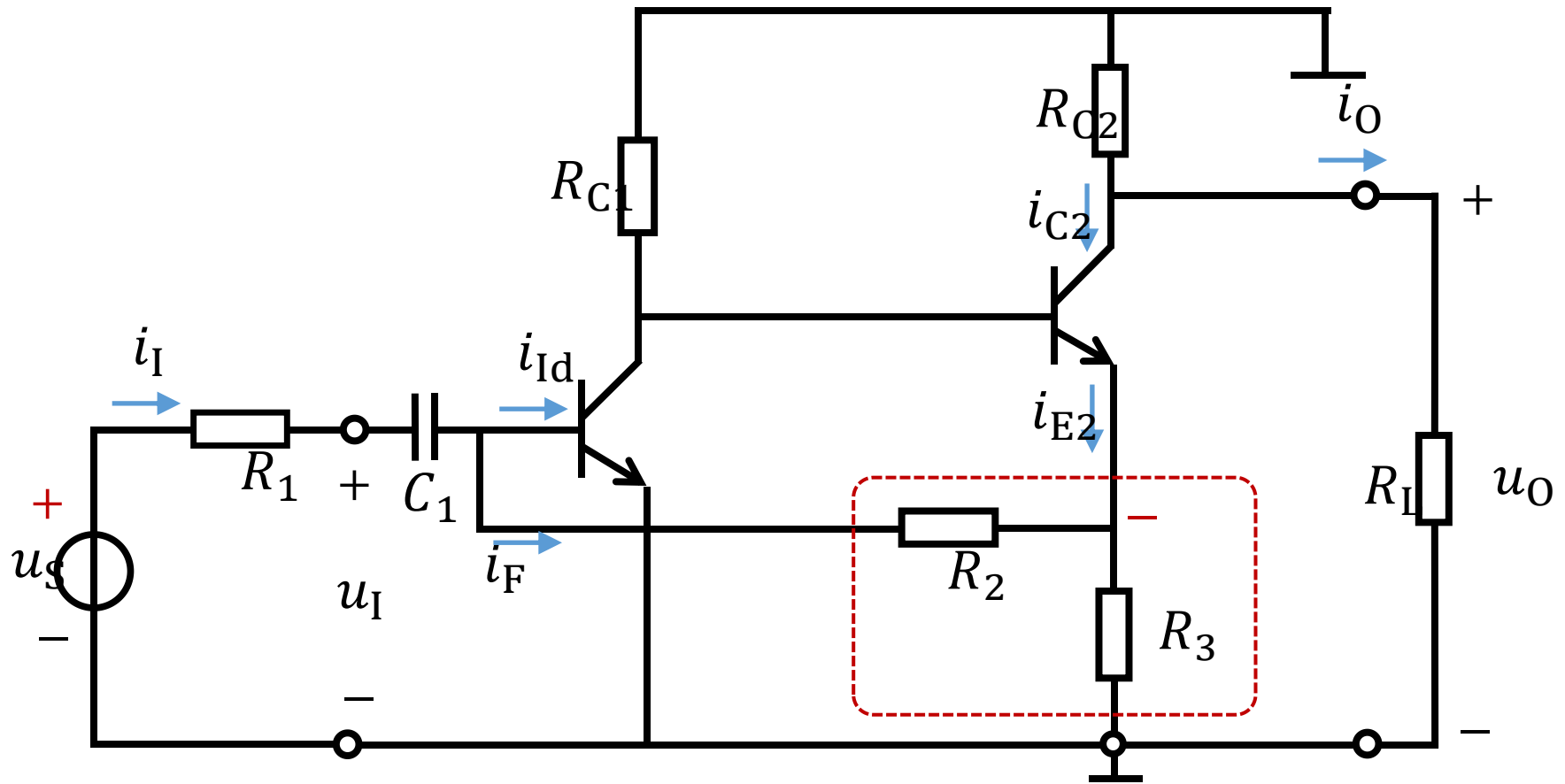


$$i_{Id} < i_I$$

Negative feedback

Example 6:

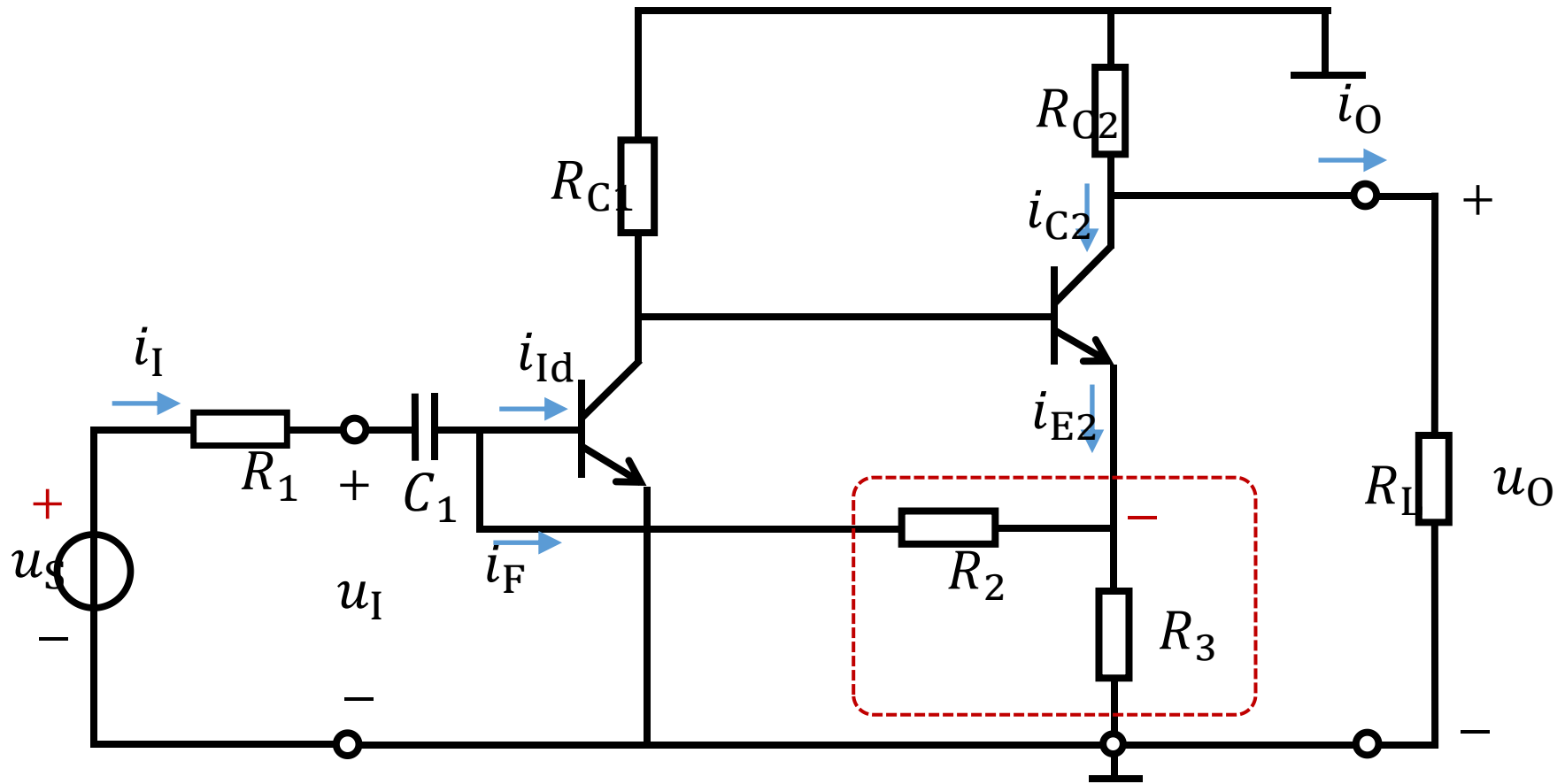
(3) Judge the type of AC negative feedback circuit



The feedback current i_F comes from i_{C2} , while not i_O

Example 6:

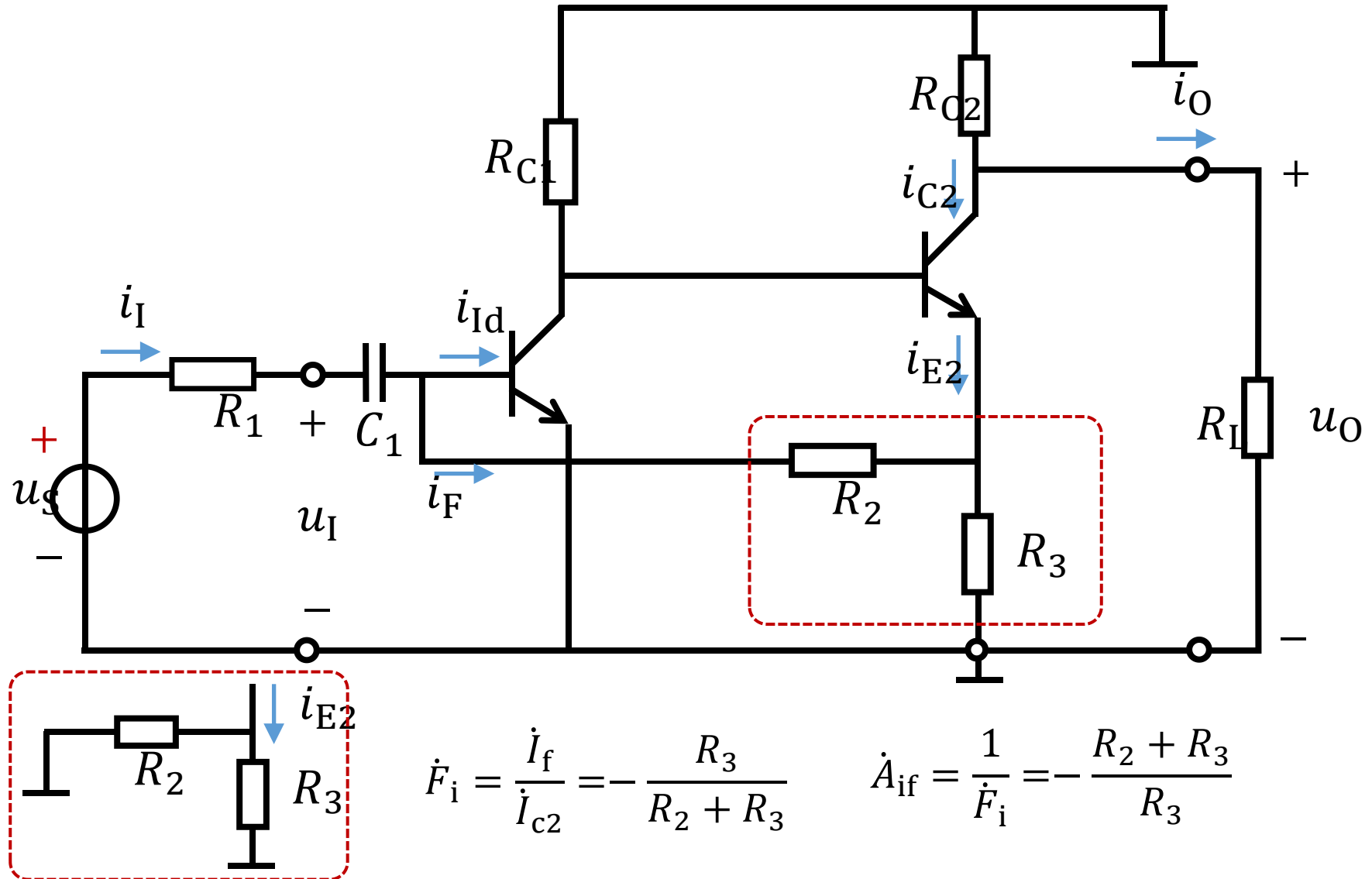
(3) Judge the type of AC negative feedback circuit



Current parallel feedback 电流并联负反馈

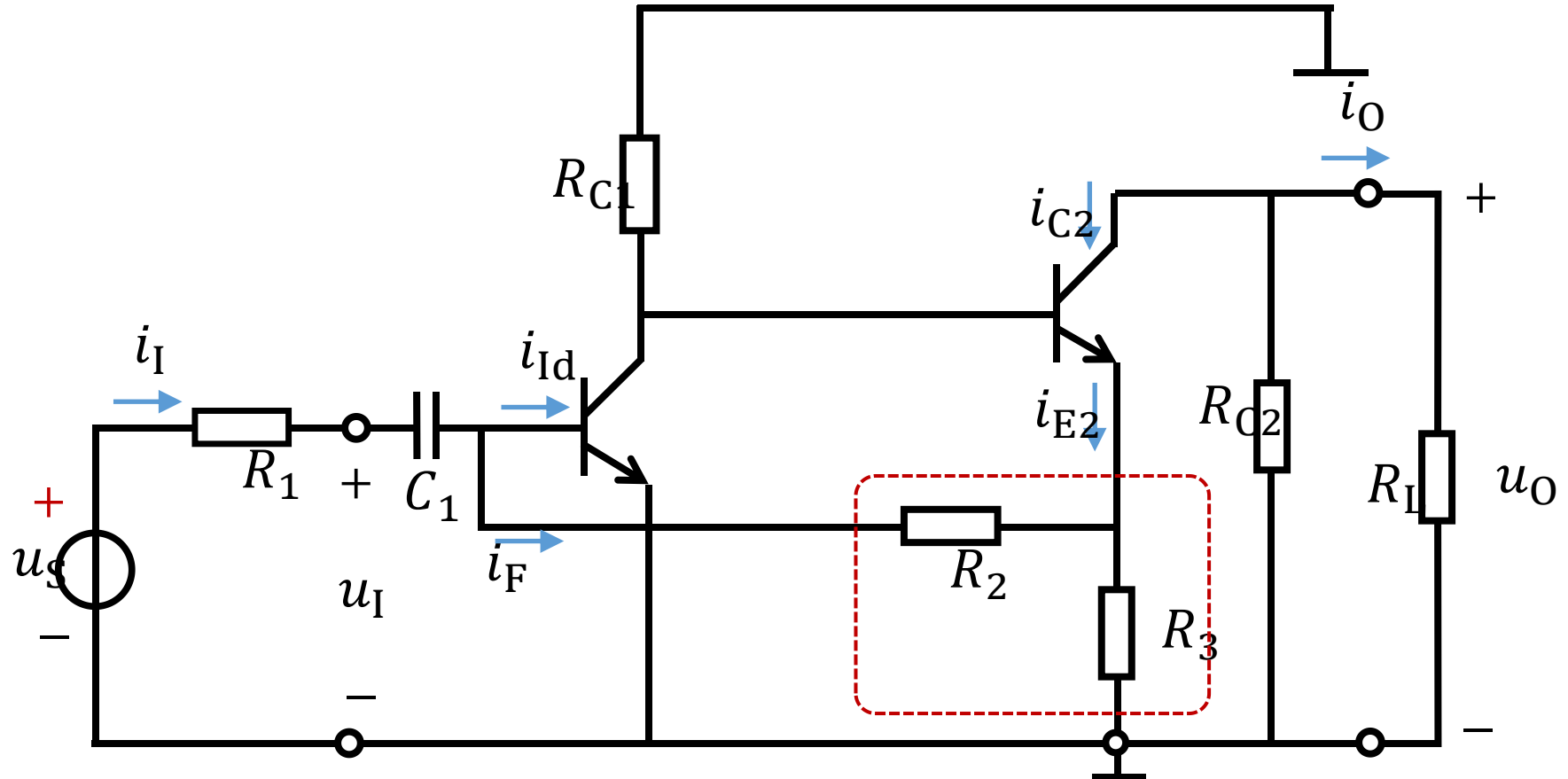
Example 6:

(4) Feedback factor and close-loop gain



Example 6:

(5) Close-loop voltage gain

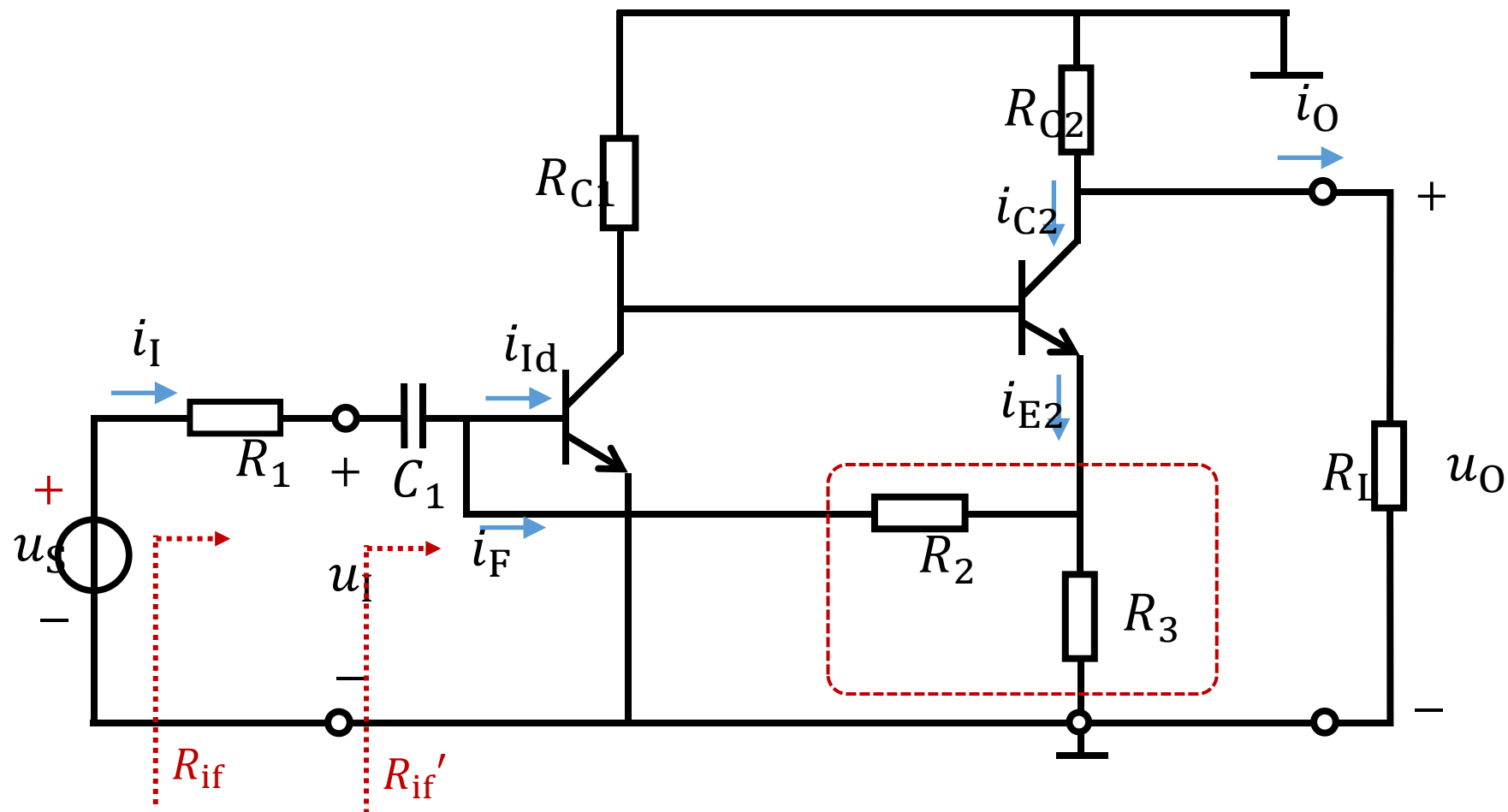


$$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_s} \quad \dot{U}_s = \dot{I}_i R_1 \approx \dot{I}_f R_1 \quad \frac{\dot{I}_f}{\dot{I}_{c2}} = -\frac{R_3}{R_2 + R_3} \quad \dot{A}_{uf} = \frac{R'_L(R_2 + R_3)}{R_1 R_3}$$

$$\dot{U}_o = -\dot{I}_{c2}(\underline{R_{C2} // R_L}) \quad R'_L$$

Example 6:

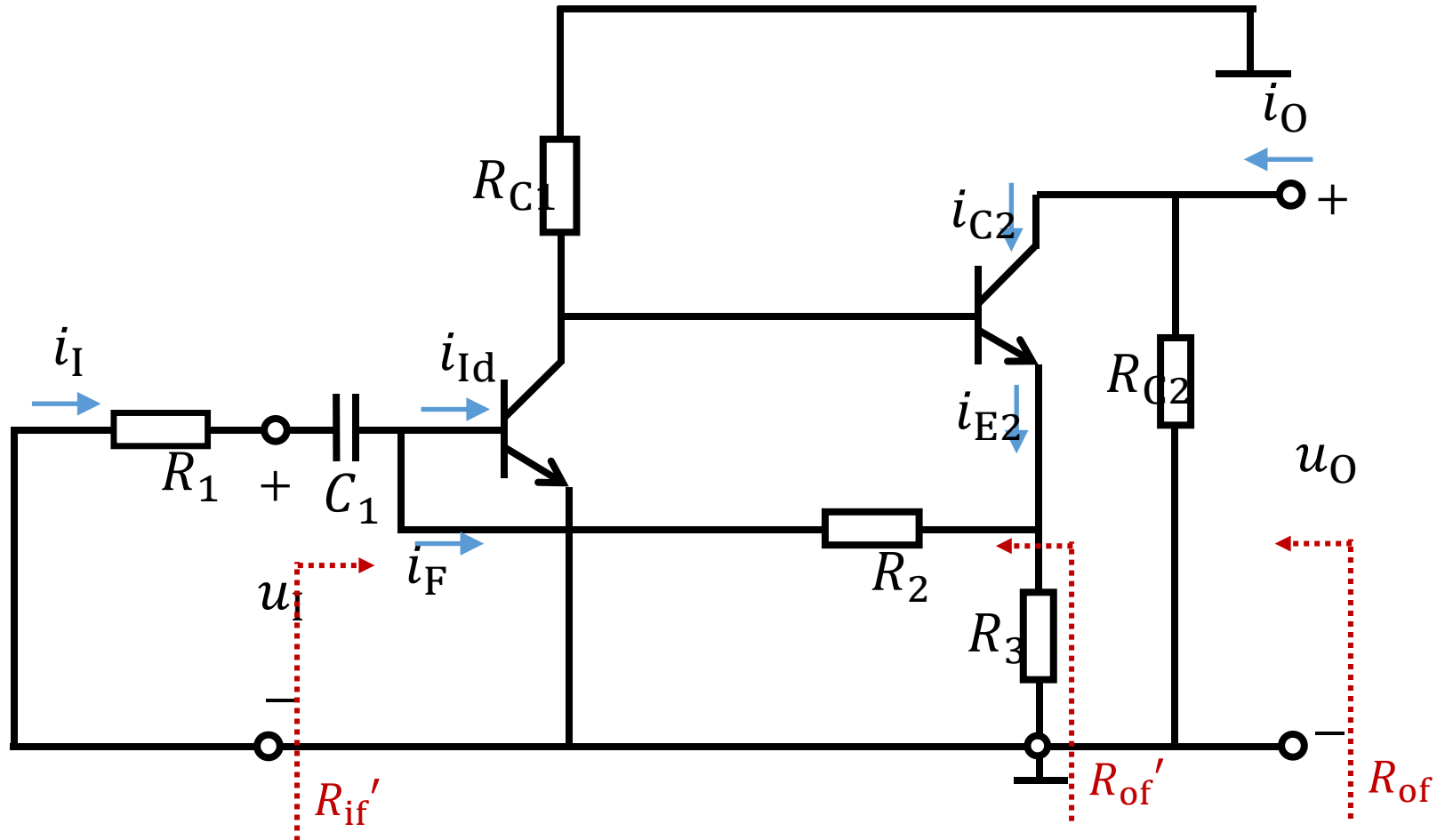
(6) Input resistance



$$R_{if} = R_1 + R_{if}' \approx R_1$$

Example 6: strong negative feedback circuit

(7) Output resistance



$$R_{of} = R_{C2} // R_{of}' \approx R_{C2}$$

Negative feedback circuit with ideal operational amplifier

理想运放的负反馈电路

An ideal case of strong negative feedback
深度负反馈的理想情况

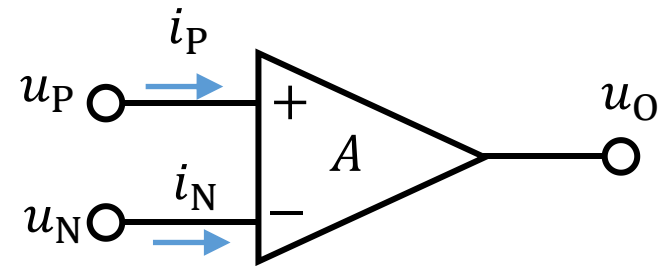
Ideal operational amplifier

Input resistance $R_i = \infty$

Output resistance $R_o = 0$

Open loop gain $A = \infty$

Upper cut-off frequency $f = \infty$



Virtual short circuit

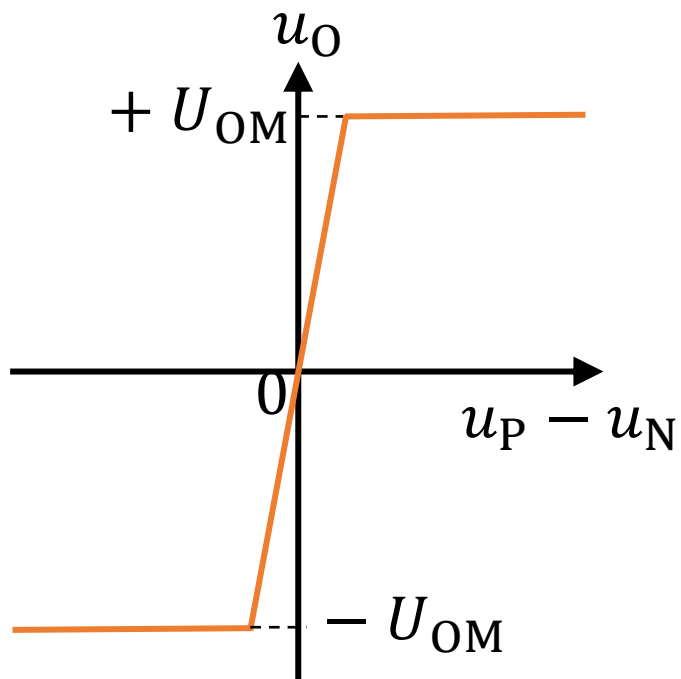
$$u_P = u_N$$

(True when in linear amplification region)

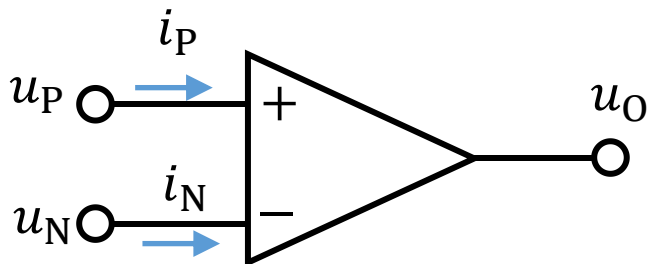
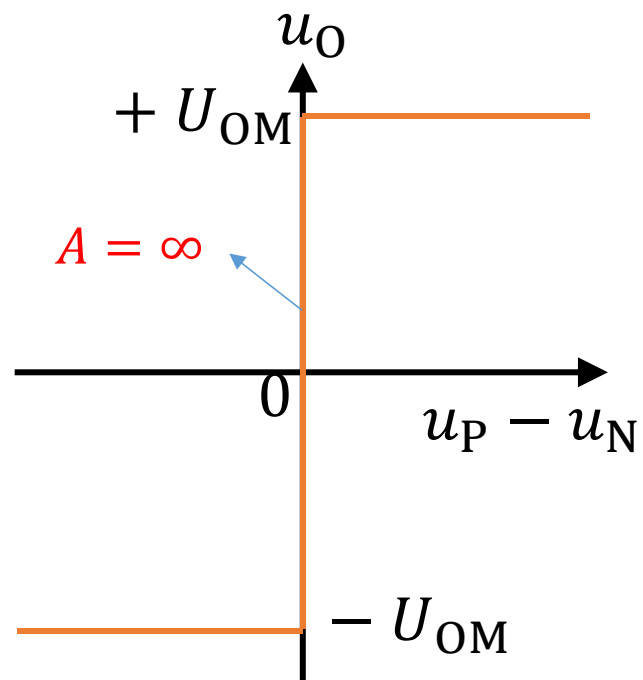
Virtual open circuit

$$i_P = i_N = 0$$

Real case



Ideal case



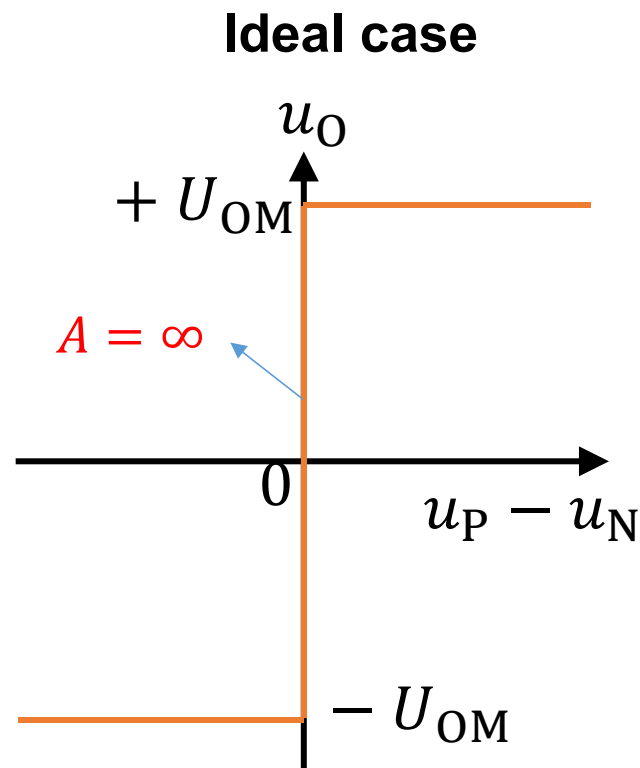
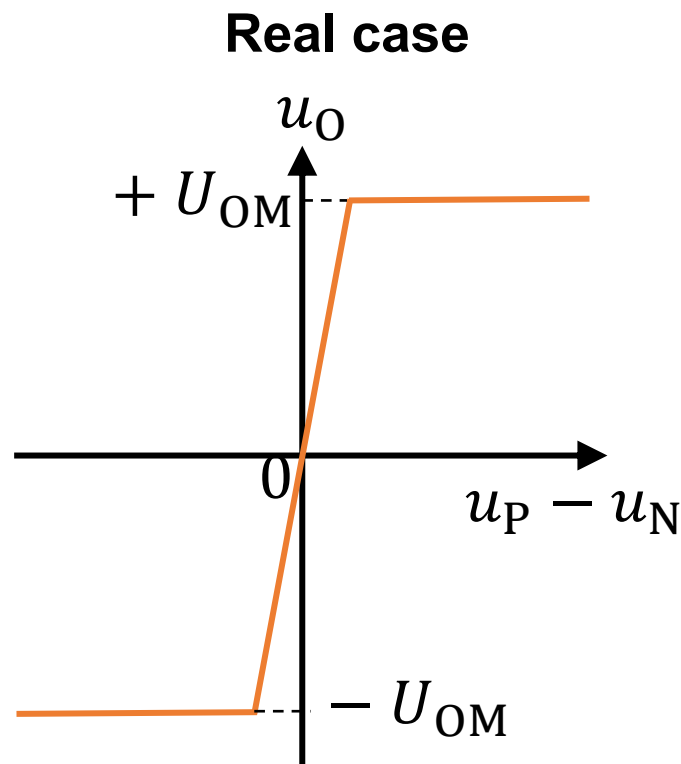
Virtual short circuit

$$u_P = u_N$$

(True when in linear amplification region)

Virtual open circuit

$$i_P = i_N = 0$$



For negative feedback circuit, the negative feedback will continuously decrease the input signal to a small value ($|u_P - u_N| = 0$ for ideal case). Hence it works in linear region.

因为负反馈电路会使净输入信号 $|u_P - u_N|$ 一直减小到很小的值（理想运放情况减小至0），所以工作在线性区。

Example 1: Voltage series negative feedback 电压串联负反馈

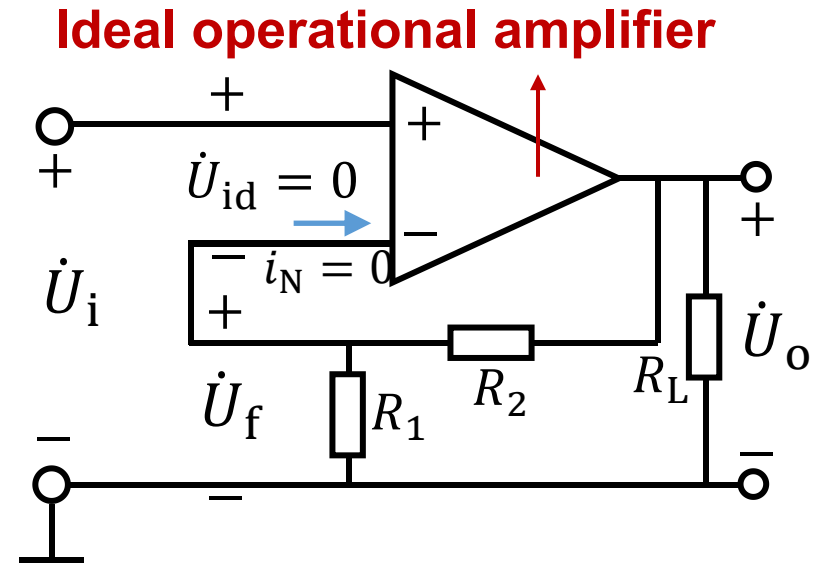
Close-loop gain

$$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_i} = \frac{\dot{U}_o}{\dot{U}_f} = \frac{1}{\dot{F}_u}$$

Strong negative feedback!

$$\dot{U}_f = \frac{R_1}{R_1 + R_2} \dot{U}_o$$

$$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_f} = \frac{R_1 + R_2}{R_1}$$



The result is the same as that using the separation method 分离法

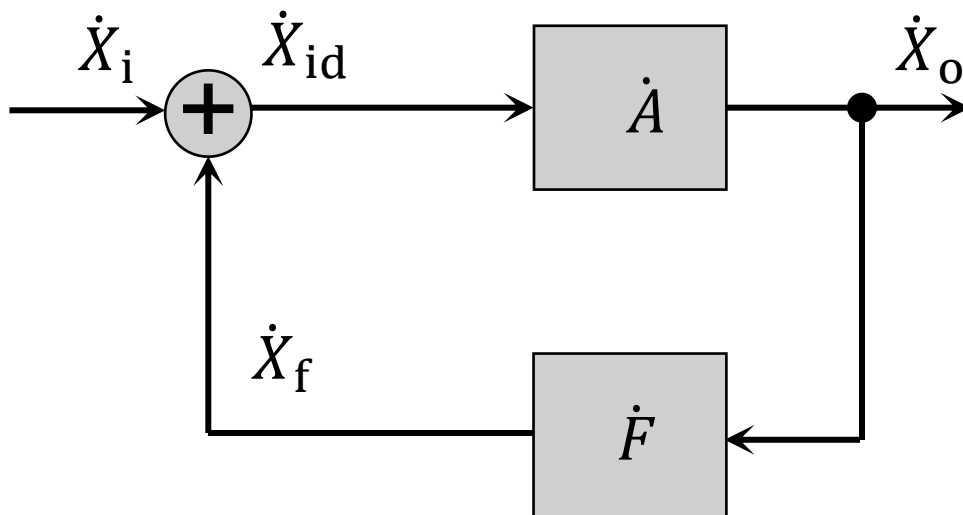
Compare

Strong negative feedback

$$\begin{cases} \dot{A}_f \approx \frac{1}{\dot{F}} \\ |\dot{A}\dot{F}| \gg 1 \end{cases} \quad \begin{cases} \dot{X}_i \approx \dot{X}_f \\ \dot{X}_{id} \approx 0 \end{cases}$$

Ideal operational amplifier

$$\begin{cases} \dot{A}_f = \frac{1}{\dot{F}} \\ \dot{A}\dot{F} \rightarrow \infty \end{cases} \quad \begin{cases} \dot{X}_i = \dot{X}_f \\ \dot{X}_{id} = 0 \end{cases}$$



That's why we can separate the feedback network from the circuit and calculate $\dot{F}$ separately.

6.5 Self-excited oscillation in negative feedback circuits

负反馈电路中的自激振荡

Close-loop gain $\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$ $\dot{X}_{id} = \dot{X}_i - \dot{X}_f$

At **midband**, negative feedback: $|1 + \dot{A}\dot{F}| > 1$.

$$\left\{ \begin{array}{l} |\dot{X}_{id}| = |\dot{X}_i| - |\dot{X}_f| \\ \varphi_A + \varphi_F = 2n\pi \end{array} \right.$$

At **low or high frequency band**, $\dot{A}$ and $\dot{F}$ are complex numbers, and $|1 + \dot{A}\dot{F}| < 1$ is possible.

Self-excited oscillation 自激振荡: $1 + \dot{A}\dot{F} = 0$

There is output AC signal when no input AC signal.

Self-excited oscillation 自激振荡: $\dot{A}\dot{F} = -1$

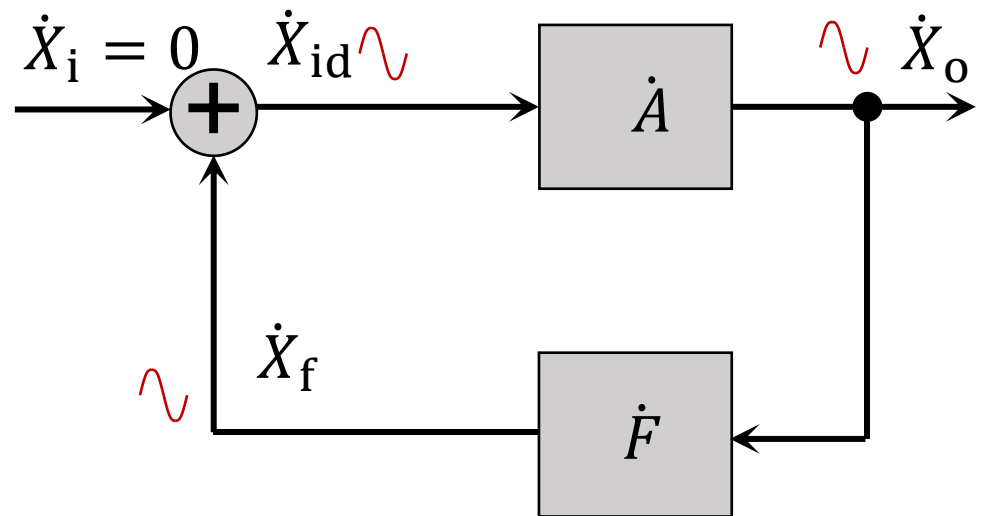
$$\begin{cases} |\dot{A}\dot{F}| = 1 \\ \varphi_A + \varphi_F = 2n\pi + \pi \end{cases}$$

$|\dot{X}_{id}| = |\dot{X}_i| + |\dot{X}_f|$: Negative feedback $\rightarrow$ Positive feedback

When $\dot{A}\dot{F} = -1$

$$\dot{X}_f = \dot{A}\dot{F}\dot{X}_{id} = -\dot{X}_{id}$$

When $\dot{X}_i = 0$,
perturbation at $\dot{X}_o$ cannot
be amplified



Start-oscillation condition 起振条件:

$$\left\{ \begin{array}{ll} |\dot{A}\dot{F}| > 1 & \text{Perturbation signal can be amplified} \\ \varphi_A + \varphi_F = 2n\pi + \pi & \text{Positive feedback} \end{array} \right.$$

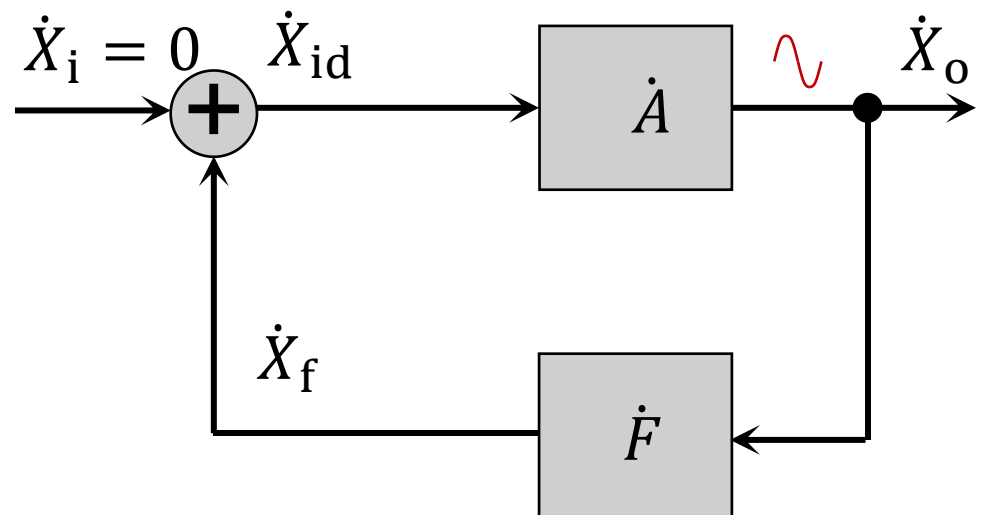
$$|\dot{X}_{id}| = |\dot{X}_i| + |\dot{X}_f|$$

When $\dot{X}_i = 0$, perturbation leads to:

$$\sim |\dot{X}_o| \uparrow \Rightarrow |\dot{X}_f|$$



$$\sim |\dot{X}_o| \uparrow\uparrow \Leftarrow |\dot{X}_{id}|$$

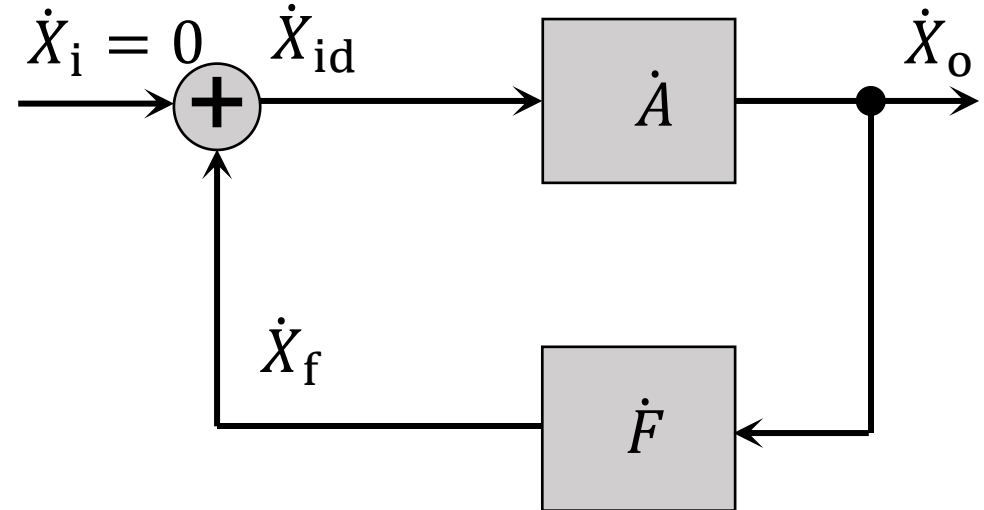


When $\dot{X}_i = 0$, perturbation leads to:

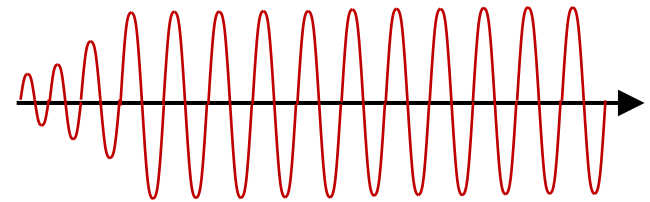
$\sim |\dot{X}_o| \uparrow \Rightarrow |\dot{X}_f|$



$\sim |\dot{X}_o| \uparrow\uparrow \Leftarrow |\dot{X}_{id}|$



When $|\dot{X}_{id}|$ is large enough, amplifier starts to enter saturation region, and $|\dot{A}|$ decrease till $|\dot{A}\dot{F}| = 1$.



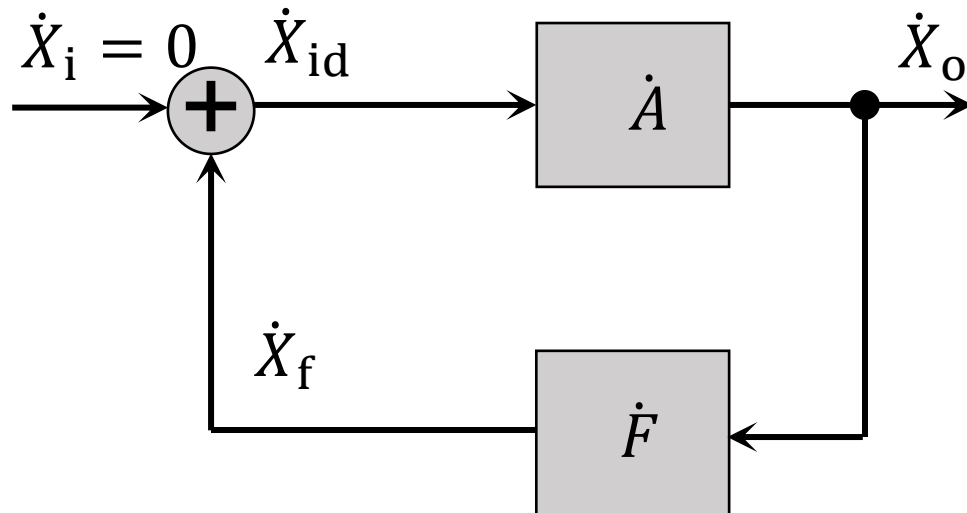
This is the self-excited oscillation

When oscillation is stable:

$$\dot{X}_f = -\dot{A}\dot{F}\dot{X}_{id}$$

$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| = 1 \\ \varphi_A + \varphi_F = 2n\pi + \pi \end{array} \right.$$

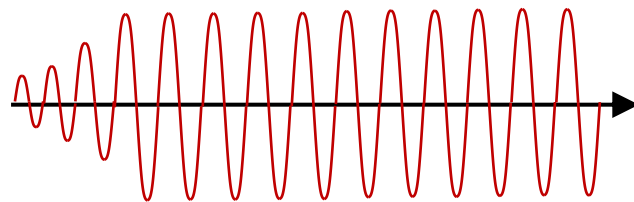
$$\dot{X}_f = \dot{X}_{id}$$



Start-oscillation condition

起振条件:

$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| > 1 \\ \varphi_A + \varphi_F = 2n\pi + \pi \end{array} \right.$$



Q1: Is there self-excited oscillation in one-stage amplifier circuit?

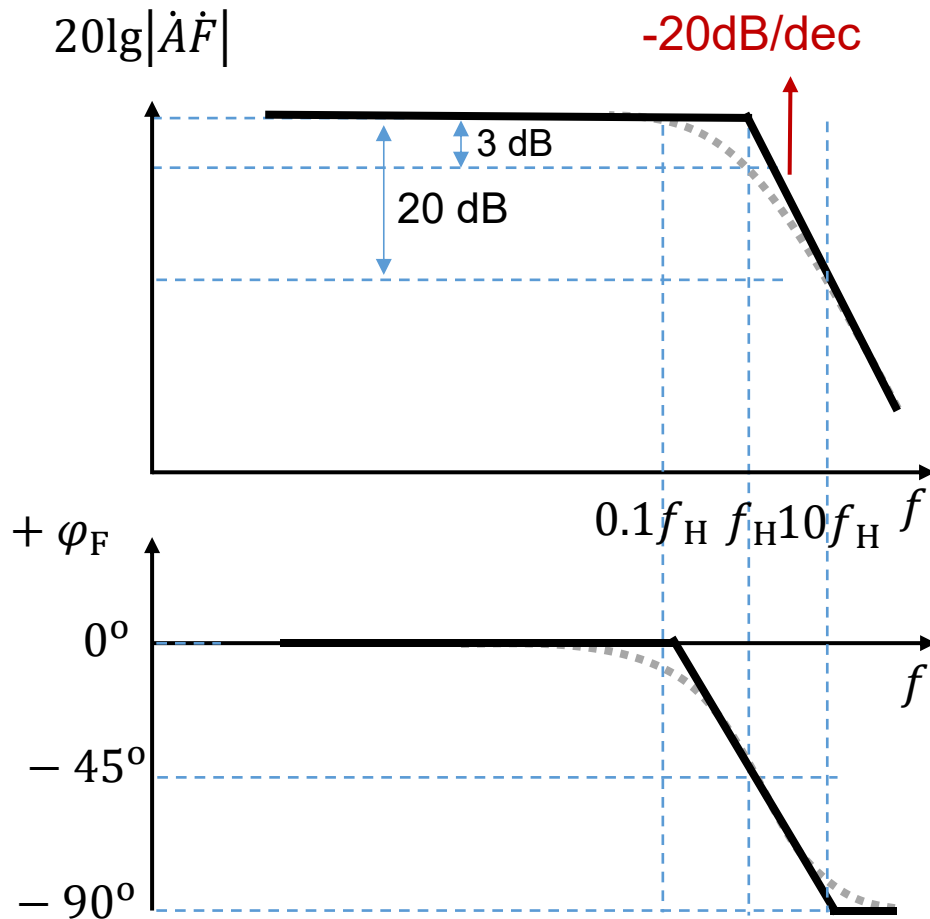
Q2: Is there self-excited oscillation in two-stage amplifier circuit?

Q3: Is there self-excited oscillation in three-stage amplifier circuit?

One-stage amplifier: high frequency region as an example

$$20\lg|\dot{A}_{\text{ush}}\dot{F}|$$
$$= 20\lg|\dot{A}_{\text{usm}}\dot{F}| - 20\lg\sqrt{1 + \frac{f^2}{f_H^2}}$$

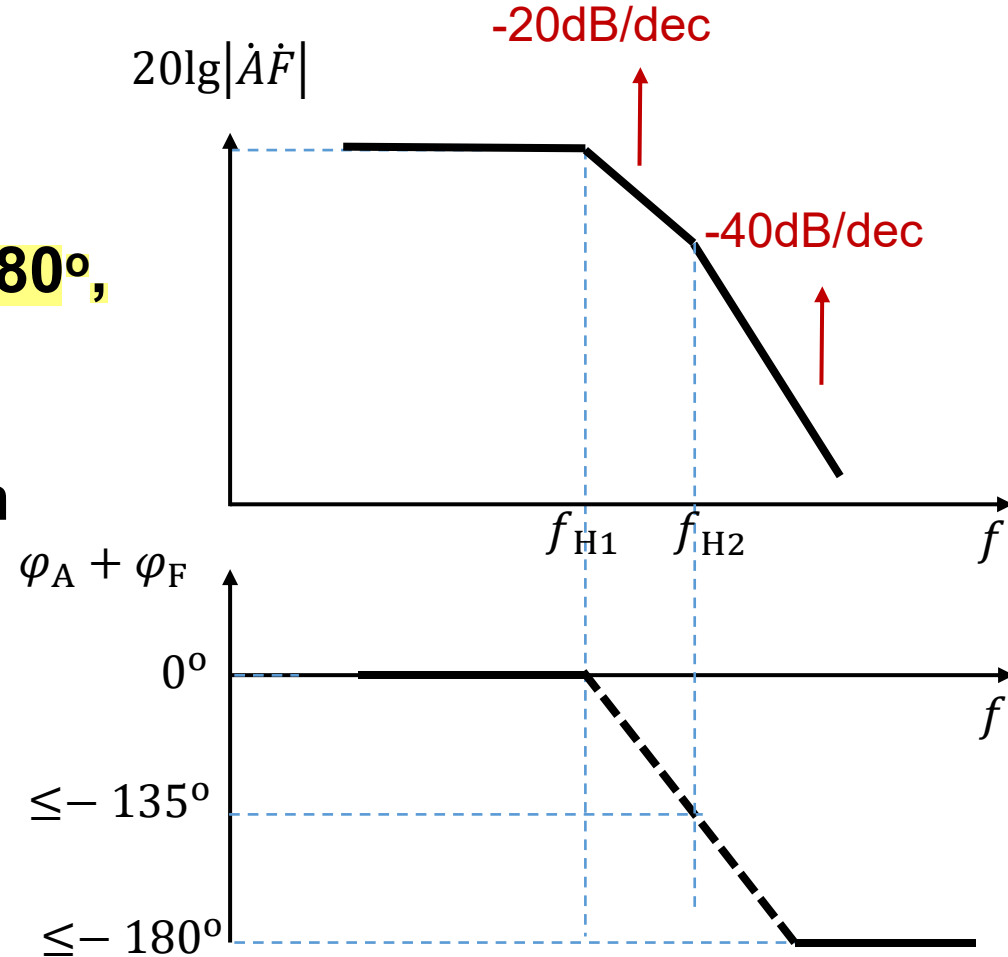
**Phase at most changes -90° ,
no self-excited oscillation**



Two-stage amplifier

Phase at most changes **-180°**,
 f_o is infinite

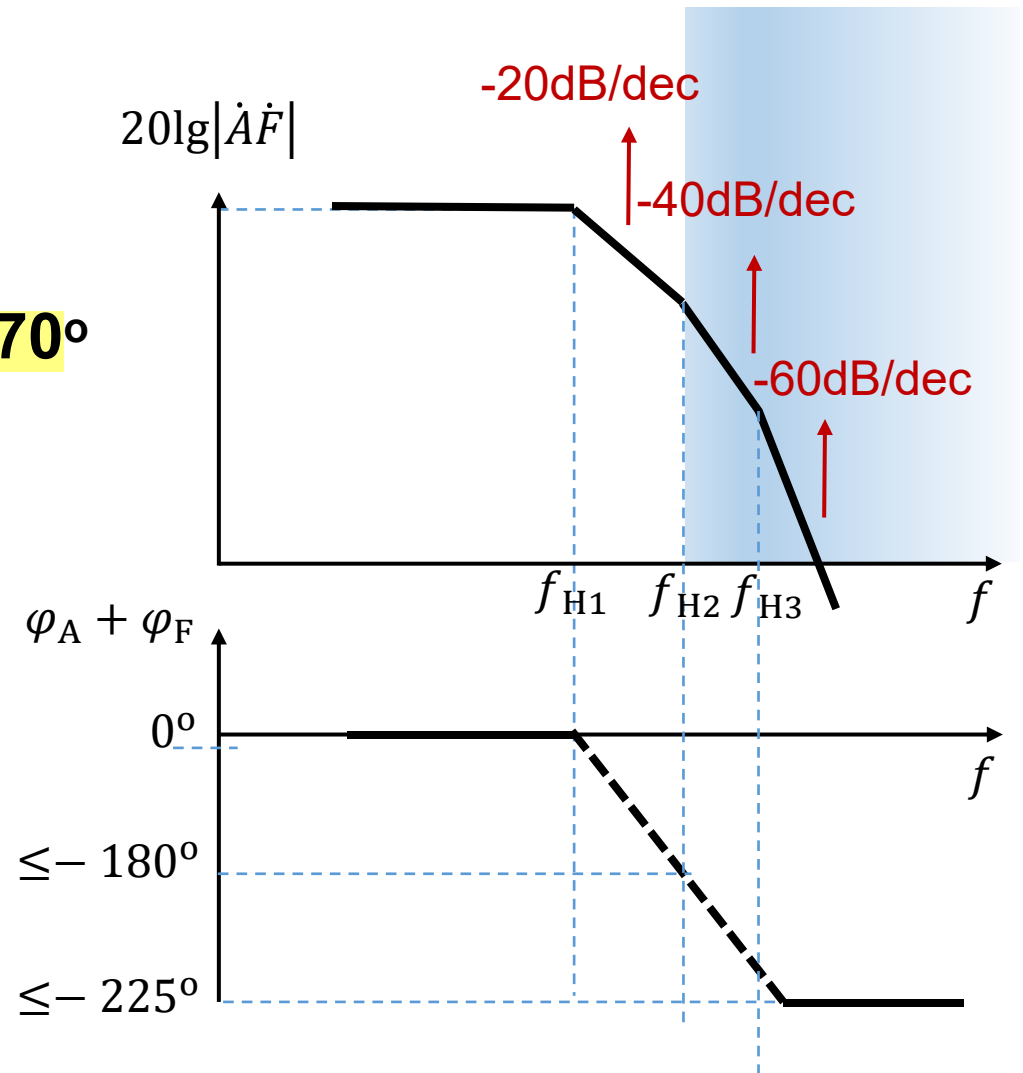
No self-excited oscillation



Three-stage amplifier

Phase at most changes **-270°**

Self-excited oscillation is possible!



Multistage amplifier with more stages is more easier to generate the self-excited oscillation (at least three stage).

Larger amount of feedback $|1 + \dot{A}\dot{F}|$, easier for self-excited oscillation

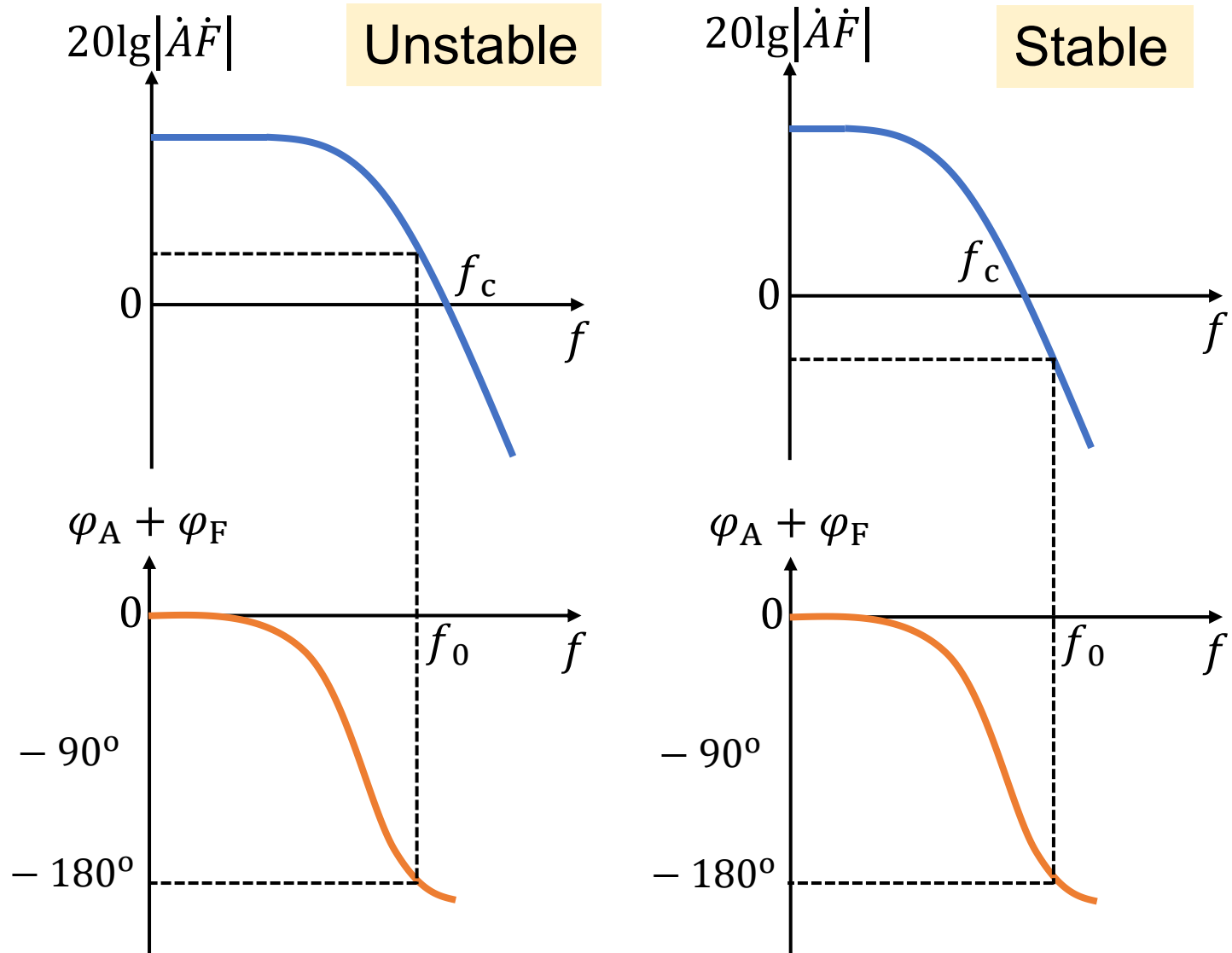
How to judge the stability of the negative feedback circuit

怎样判断负反馈电路的稳定性

(1) Find the critical frequency f_0 for $\varphi_A + \varphi_F = -180^\circ$.

$\left\{ \begin{array}{l} \text{If } |\dot{A}\dot{F}| \geq 1 \text{ at } f_0, \text{ the circuit is unstable} \\ \text{If } |\dot{A}\dot{F}| < 1 \text{ at } f_0, \text{ the circuit is stable} \end{array} \right.$

$$|\dot{A}\dot{F}| \geq 1 \quad \rightarrow \quad 20\lg|\dot{A}\dot{F}| \geq 0\text{dB}$$



How to judge the stability of the negative feedback circuit

怎样判断负反馈电路的稳定性

(2) Find the critical frequency f_c for $|\dot{A}\dot{F}| = 1$ and critical frequency f_0 for $\varphi_A + \varphi_F = -180^\circ$.

{ If $f_c \geq f_0$, the circuit is unstable
If $f_c < f_0$, the circuit is stable

Gain and phase stability margin 幅值和相位稳定裕度

Although $f_c < f_0$, the circuit is stable,
we usually require the circuit:

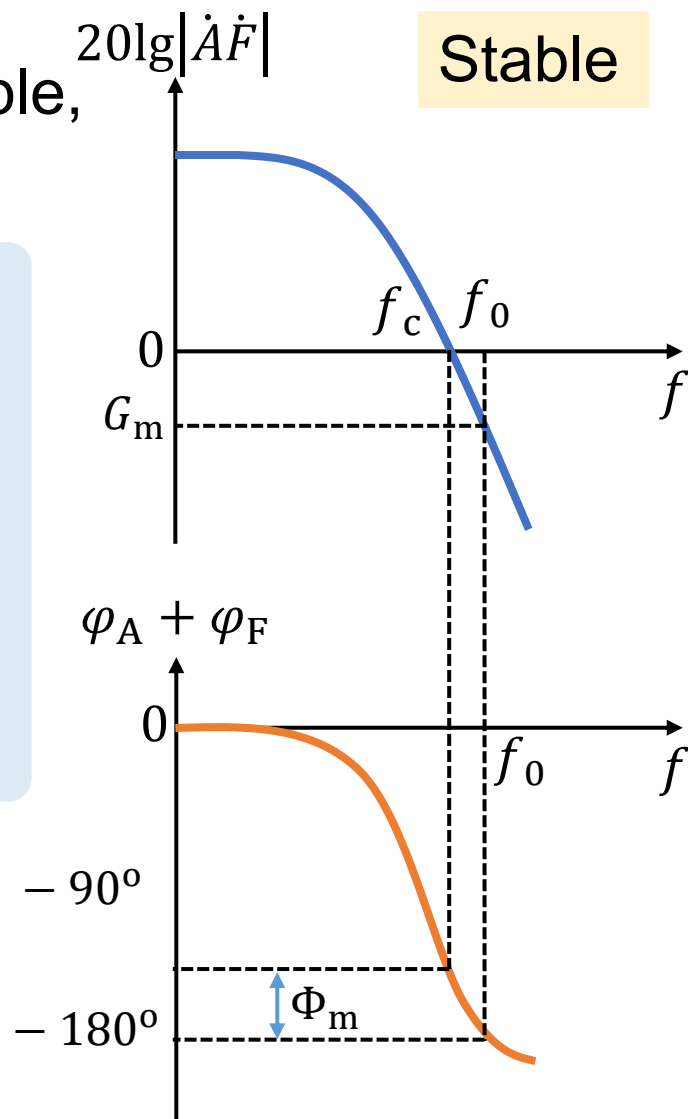
Gain stability margin $G_m(\text{dB})$

$$G_m < -10 \text{ dB}$$

Phase stability margin Φ_m

$$\Phi_m > 45^\circ$$

When $G_m < -10 \text{ dB}$ and $\Phi_m > 45^\circ$,
the circuit is very reliable.



Gain and phase stability margin 幅值和相位稳定裕度

Although $f_c < f_0$, the circuit is stable,
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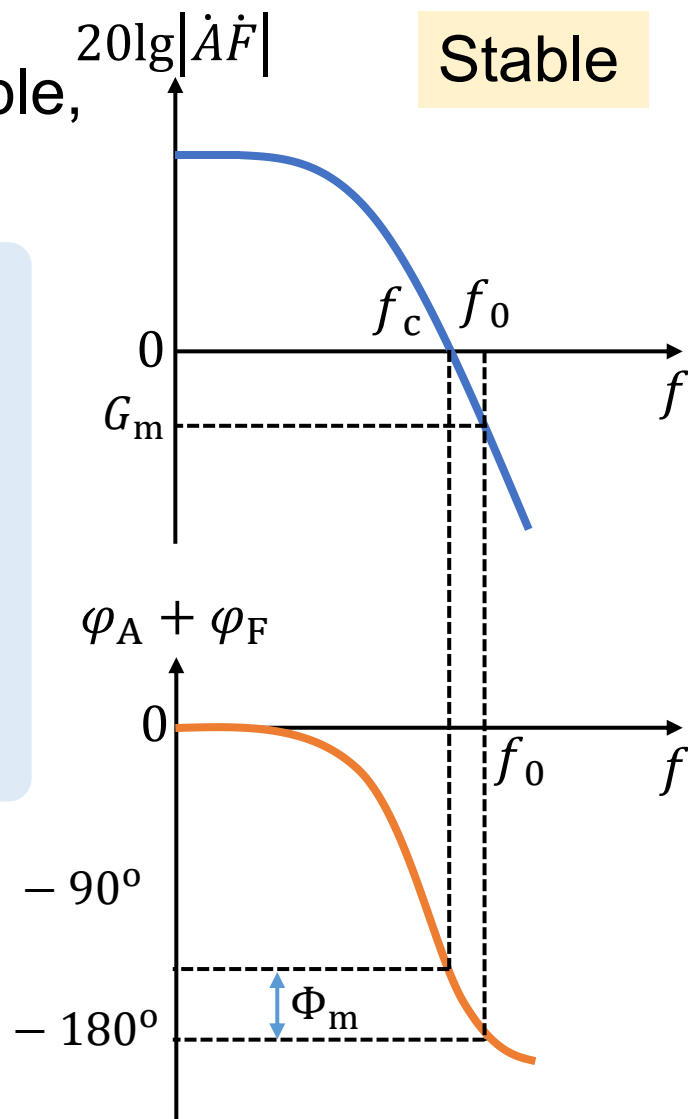
Gain stability margin $G_m(\text{dB})$

$$G_m < -10 \text{ dB}$$

Phase stability margin Φ_m

$$\Phi_m > 45^\circ$$

When $G_m < -10 \text{ dB}$ and $\Phi_m > 45^\circ$,
the circuit is very reliable.



Methods to eliminate the self-excited oscillation

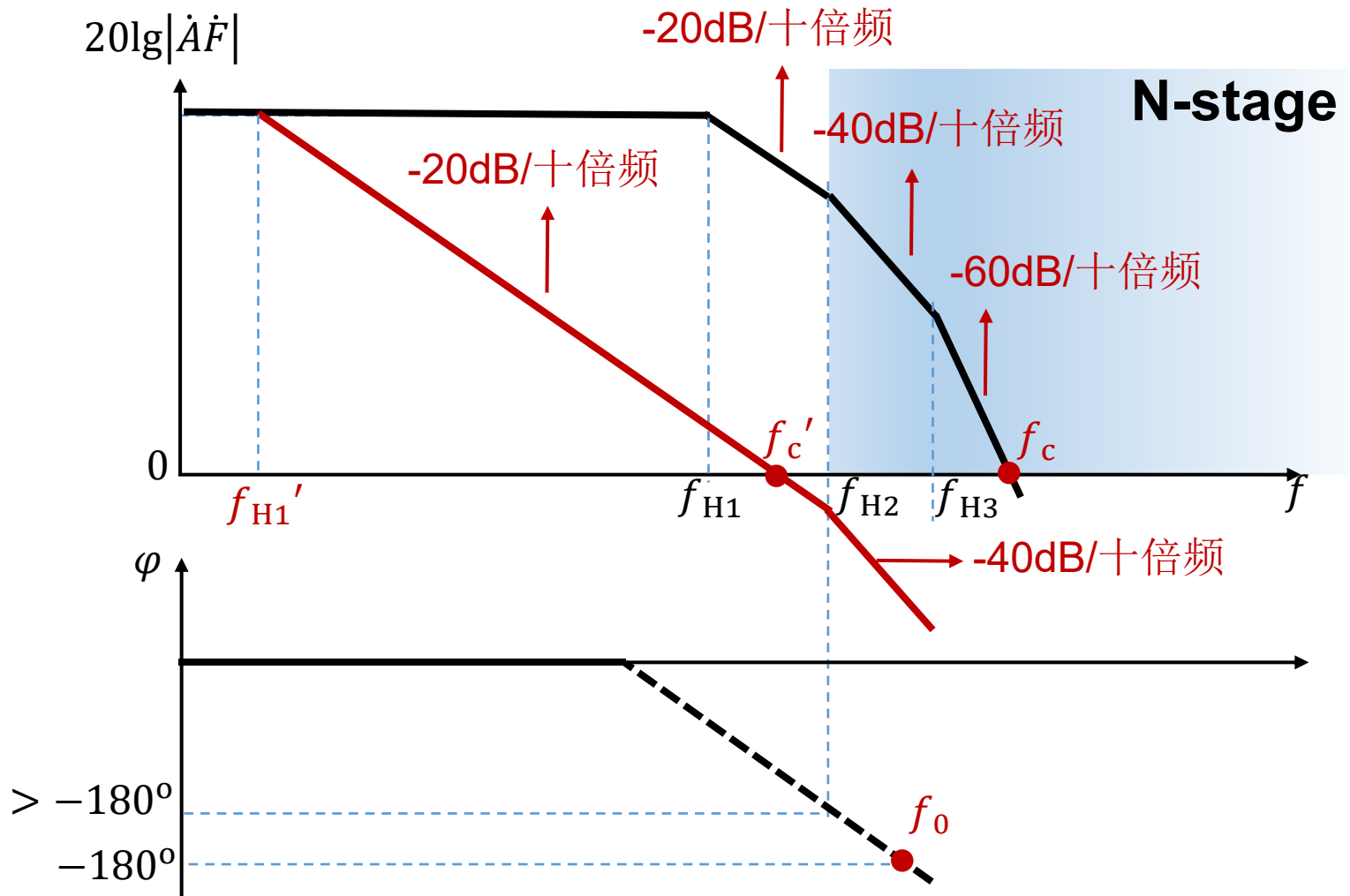
消除自激振荡的方法

Compensation 补偿 $f_c < f_0$

Add resistor and capacitor in the amplifier circuit, to increase the gain and phase stability margin.

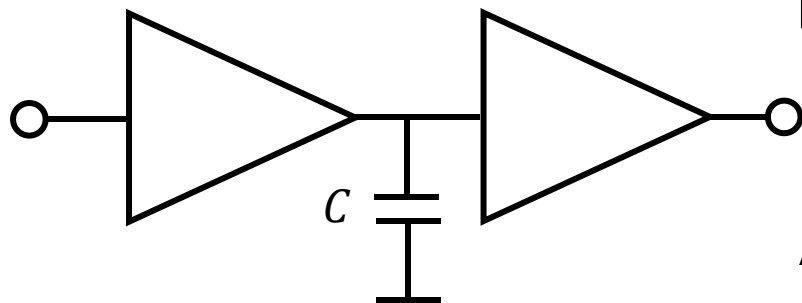
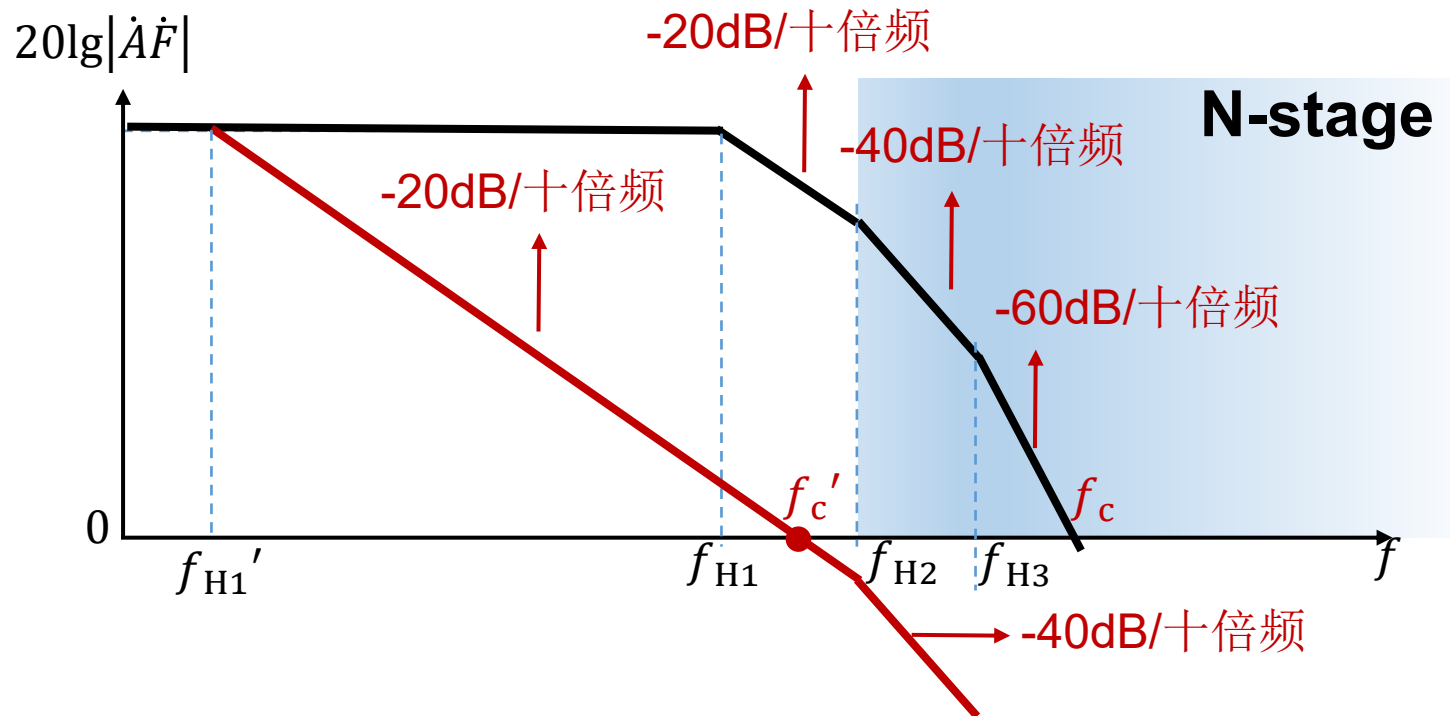
- Lag compensation 滞后补偿
- Lead compensation 超前补偿

Lag compensation 滞后补偿



$f_c' \leq f_0$: No self-excited oscillation

Lag compensation 滞后补偿



Before compensation: $f_H = \frac{1}{2\pi RC_1}$

After compensation: $f_H = \frac{1}{2\pi R(C_1 + C)}$

Homework 7: Refer to blackboard and QQ group

Deadline: 2025-05-21 11:59pm